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# Pension Liquidity Risk

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Pension funds use interest rate swaps to hedge the interest rate risk arising from their liabilities. Analyzing regulatory data on Dutch pension funds, we show that pension funds with worse funding ratios, indicating greater fragility, use swaps more aggressively. These swap positions expose pension funds to the risk of margin calls, which can exceed 6% of their total assets, when interest rates rise. Pension funds respond to realized margin calls by selling safe government bonds with medium-term maturities. This procyclical selling behavior adversely affects the prices of the sold bonds and thereby exposes pension funds to market liquidity risk. (*JEL* E43, G12, G18)

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Authors have furnished an Internet Appendix, which is available on the Oxford University Press Web site next to the link to the final published paper online.

Pension funds are huge institutional investors with total assets under management (AUM) that often exceed the gross domestic product (GDP) of their home countries (OECD 2022). Given the sheer size of the pension sector, pension funds' hedging decisions have received substantial attention in the academic literature, highlighting their demand for long-dated government bonds and interest rate swaps (e.g., Greenwood and Vayanos 2010; Greenwood and Vissing-Jorgensen 2018; Klingler and Sundaresan 2019; Jansen 2025). By contrast, little is known about the adverse effects of pension funds' hedging strategies using derivatives. For instance, when a pension fund receives the fixed rate in a long-dated interest rate swap, an increase in interest rates requires the fund to post additional collateral, typically in the form of cash. Such additional margin requirements, or margin calls, can force the pension fund to sell part of its asset holdings at depressed prices, thereby exposing it to market liquidity risk (Brunnermeier and Pedersen 2009).

In this paper, we use unique regulatory data on Dutch pension funds to examine how hedging with interest rate swaps exposes pension funds to liquidity risk. We begin by documenting that pension funds with worse funding ratios (i.e., ratios between the present value of assets and liabilities) use interest rate swaps more aggressively. These swap positions reduce interest rate risk but introduce the risk of substantial margin calls that can exceed 6% of the pension funds' total assets when interest rates rise. Turning to asset allocation, we find that pension funds with large swap positions reduce their bond holdings in response to rising swap rates. When margin calls materialize during episodes of significant rate increases, pension funds primarily sell their safe euro-denominated government bonds with medium-term maturities between 1 and 7 years, which are their safest and most liquid assets. These sales generate adverse price effects. Thus, due to margin calls from derivatives positions, pension funds' hedging decisions can even impact the medium-term segment of the yield curve.

Our analysis draws on detailed regulatory data covering the Dutch pension system from 2012 to 2022. With a total of $2.04 trillion in AUM as of 2021, the Dutch pension system represents 53% of all pension assets in the euro area (ECB 2022) and corresponds to 209.5% of the Dutch GDP (OECD 2022). Additionally, the Dutch pension system is an ideal laboratory for investigating the link between the use of derivatives and asset holdings, given our access to detailed information on pension funds' liabilities, asset holdings, and derivative positions.

We begin our analysis by documenting three stylized facts. First, Dutch pension funds predominantly receive the fixed rate in long-dated swaps,

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and their net swap exposures exceed those of all other Dutch counterparties at maturities beyond 5 years. Second, to hedge the duration risk of their liabilities, pension funds rely mainly on swaps rather than bonds.[1] Based on the dollar duration of the swap and bond portfolio, scaled by AUM, we estimate that approximately 60% of their total portfolio duration comes from swap positions.[2] Third, realized margin calls are substantial, averaging 3% of AUM during periods of interest rate surges and frequently exceeding the cash buffers of affected pension funds. In extreme cases, margin calls exceed 6% of AUM. For the majority of pension funds, these margin calls also exceed the improvements in funding ratios that result from rising interest rates.

We next derive and test three hypotheses. Our first hypothesis is that pension funds with lower funding ratios hold larger positions in interest rate swaps. The rationale for this hypothesis is that pension funds cannot use direct leverage to increase their bond positions. Instead, they can use interest rate swaps to hedge rate movements while maintaining their strategic asset allocations.[3] Consistent with this hypothesis, we show that pension funds with lower funding ratios indeed hold larger swap positions. These funds also hold less cash. Hence, their larger swap positions expose them to the risk of forced portfolio liquidations when interest rates—and thus the margin requirements on their swap positions—rise. This liquidity risk is more pronounced for pension funds with a lower funding ratio.

Our second hypothesis is that pension funds respond to rising swap rates by selling their most liquid assets. To test this hypothesis, we study how changes in swap rates affect pension funds' asset allocations. We find that pension funds reduce their bond holdings when swap rates rise. These reductions contrast with strategic rebalancing, which would imply purchasing bonds when prices fall. To connect the selling of fixed income securities to margin calls, we show that the reduction in bond holdings is driven by pension funds with longer swap durations. The cross-sectional link between increasing swap rates and pension funds' reductions in fixed income holdings mitigates the concern that our findings are driven by the endogenous determination of interest rates. To further alleviate such concerns, we show that the results are robust to instrumenting the 20-year swap rate with the effect of monetary policy surprises (MPSs) on the 20-year euro Overnight Indexed Swap (OIS) rate. In addition, funds with long bond durations instead of long swap durations do not sell bonds when rates

[1] Throughout, we define "duration" as the price sensitivity to interest rate risk because pension regulation imposes zero duration on equities.

[2] This fraction equals 46% when using a value-weighted average based on pension fund size.

[3] Pension funds therefore use swaps to reduce their interest rate risk exposure while maintaining risky asset allocations. We refer to this type of swap usage as hedging.



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increase, making it unlikely that the observed pattern is driven by changes in interest rate expectations or anticipated bond returns.

Turning to the security level, our third hypothesis is that pension funds sell their safest and most liquid securities in response to realized margin calls. To test this hypothesis, we examine security-level bond holdings during months with interest rate surges. Pension funds first liquidate cash-like assets such as money market funds (MMFs), but these holdings are insufficient to meet typical margin call sizes.[4] The second-most liquid assets of Dutch pension funds are German and Dutch government bonds (the only euro-denominated government bonds with a triple-A rating from all major rating agencies over our sample period), and in line with our hypothesis, pension funds sell these government bonds in response to margin calls. Specifically, the selling of safe government bonds is concentrated in medium-term maturities (tenors between 1 and 7 years), which we show are more liquid than longer-dated bonds. These results also help rule out the alternative explanation that pension funds sell bonds to manage their duration gap, whereby rising rates mechanically reduce the gap between liability and asset duration and may prompt bond sales to keep this gap stable. Such duration rebalancing would primarily imply the selling of long-term bonds rather than the medium-term bonds we observe. Moreover, bond sales are largest among funds with longer swap durations, which have smaller duration gaps and therefore the least need for duration rebalancing.

Because pension funds are major investors in government bond markets, it is plausible that these forced sales have an adverse price impact. To examine this possibility, we construct a bond-level proxy that captures liquidity risk arising from margin calls. This measure is the average of the pension funds' swap duration, weighted by the size of the funds' positions in that bond. We then examine the link between price changes and the lagged liquidity risk measure. A 1-standard-deviation increase in this measure reduces medium-term bond returns by 11 basis points, or 17% of the average return during periods of interest rate surges. Hence, margin calls from derivatives positions expose pension funds to market liquidity risk. Moreover, after selling the medium-term bonds, pension funds eventually need to rebalance their portfolios to comply with their strategic asset allocation. As such, our estimates are a lower bound on the total transaction costs faced by pension funds.

We conclude our price impact analysis by applying the granular instrumental variables (GIV) approach of Gabaix and Koijen (2022, 2024) and quantify the aggregate price impact of pension fund sales. Our estimates

[4] This finding aligns with Ghio et al. (2023), who show that MMF outflows in March 2020 were driven by investors facing margin calls. Stress tests also reveal that pension funds consider the sales of MMF shares as a key source to obtain liquidity (DNB and AFM 2024).



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suggest that when the 3-year safe euro yield rose by 87 basis points in August 2022, pension fund sales driven by margin calls contributed 10.4 basis points, or 12% of the total increase.

Our study reveals that hedging with interest rate swaps exposes pension funds to liquidity risk due to margin calls, challenging the traditional view that pension funds stabilize financial markets because of their long investment horizons (e.g., Grinblatt and Keloharju 2000; Timmer 2018; Chodorow-Reich, Ghent, and Haddad 2021; Coppola 2025). The liquidity risk highlighted in our study arises as a consequence of a combination of two financial regulations designed to enhance the stability of the financial system. First, regulators incentivize pension funds to hedge their interest rate risk exposure, and pension funds respond to these incentives by using interest rate swaps.[5] Second, stricter margin requirements and mandatory central clearing of these over-the-counter (OTC) derivatives increase liquidity risk for swap users via margin calls. Importantly, our paper does not challenge the view that pension funds have a stable investor base, but it demonstrates that margin calls from derivatives positions introduce another type of liquidity risk. This pension liquidity risk manifests when interest rates surge, and not necessarily during times of financial distress.

These issues extend beyond the Dutch pension system. Pension funds in the United States, Canada, the United Kingdom, and Japan also use interest rate swaps for hedging (e.g., Klingler and Sundaresan 2019; IMF 2025), and recent episodes, such as the 2022 U.K. pension crisis, demonstrate that margin call–driven fire sales can destabilize sovereign bond markets (U.K. Parliament 2022; IMF 2023). Similar concerns apply to insurance companies in Europe and the United States, which operate under regulatory frameworks that similarly combine capital constraints with mandatory collateral requirements.

Our paper contributes to three strands of literature. First, we contribute to the extensive literature that examines the risk management and hedging behavior of pension funds and insurance companies. The link between regulatory constraints, hedging, and asset allocation decisions of pension funds and insurance companies has been studied by, among others, Adams and Smith (2009); Ellul, Jotikasthira, and Lundblad (2011); Andonov, Bauer, and Cremers (2017); Ge and Weisbach (2021); Koijen and Yogo (2021); Koijen and Yogo (2022); Ellul et al. (2022); and Sen (2023). For instance, Sen (2023) shows that regulatory incentives for hedging the risks of variable annuities lead insurance companies to use derivative contracts. We contribute by highlighting a previously overlooked feedback channel: hedging with interest rate swaps exposes pension funds to liquidity risk

[5] It is important to acknowledge that a low funding ratio might be a result of more risk taking. However, the current regulatory framework strengthens the incentives of underfunded pension funds to hedge interest rate risk, and swaps provide an effective way to do so while maintaining strategic asset allocations.



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through margin calls, which can trigger forced sales of safe assets. Our results also extend the literature on pension funds' price impact on long-dated securities (e.g., Greenwood and Vayanos 2010; Domanski, Shin, and Sushko 2017; Greenwood and Vissing-Jorgensen 2018; Klingler and Sundaresan 2019; Khetan et al. 2023; Jansen 2025) by showing that such sales can also significantly affect the medium-term segment of the yield curve.

Second, our results on safe asset sales are related to the literature on (reverse) flights to liquidity (e.g., Ben-David, Franzoni, and Moussawi 2012; Manconi, Massa, and Yasuda 2012; Czech et al. 2021; Jiang et al. 2022; Ma, Xiao, and Zeng 2022; Alfaro et al. 2024). These studies show that investors facing liquidity needs systematically sell their most liquid assets first, especially during stress episodes. Our study adds to this literature by demonstrating that the pecking order of selling the most liquid assets first is not confined to a specific episode or an extreme crisis. In response to significant increases in interest rates, pension funds systematically and procyclically sell safe and liquid bonds to meet margin calls from their swap positions.

Third, we contribute to the broader literature on uninformed demand shocks and price impact (e.g., Shleifer 1986; Wurgler and Zhuravskaya 2002; Greenwood 2005; Chang, Hong, and Liskovich 2015; Koijen and Yogo 2019). In particular, Da et al. (2018) and Aldunate et al. (2025) show that frequent and uninformed reallocations between equity and bond funds by Chilean pension investors generate significant price pressure in the Chilean stock and foreign exchange markets. We contribute to this literature by showing that margin calls from derivative positions are another source of uninformed demand shocks that result in spillover effects in financial markets.

Additionally, Pinter (2023) and Pinter, Siriwardane, and Walker (2025) offer a microstructure analysis of the liquidity dry-ups during the U.K. pension crisis of late 2022. While these papers study the unfolding of one margin call episode, our study examines the entire causal chain leading to margin calls. We show that regulatory constraints incentivize swap usage, which then exposes pension funds to margin call risk and forced asset sales when interest rates rise. While Pinter, Siriwardane, and Walker (2025) find that the U.K. fire sale was amplified by balance-sheet segmentation between liability-driven investment (LDI) funds and their pension fund investors, we find that pension liquidity risk can arise even in the absence of these frictions. Using interest rate swaps allows pension funds to lower their interest rate risk and comply with regulation while maintaining their optimal asset allocation. This increases their expected returns compared to holding more bonds, but comes at the risk of potential margin calls when interest rates suddenly rise.

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## 1. Background and Stylized Facts

In this section, we first describe key institutional features of the Dutch pension system and the margin requirements on interest rate swaps, then introduce our data and present three stylized facts that motivate our analysis.

### 1.1 The Dutch pension system

There are three types of Dutch pension funds: (i) corporate pension funds, (ii) industry-wide pension funds, and (iii) professional-group pension funds.[6] Unlike in other countries, such as the United States, all three types operate under the same regulatory framework. This regulatory homogeneity is essential for our empirical analysis, in which we pool all three types of funds.

**1.1.1 Pension regulation.** To assess the financial health of a pension fund, regulators focus on two key indicators. First, the funding ratio measures the ratio of pension assets to pension liabilities. The numerator is the market value of the pension assets, and the denominator is the present value of the pension liabilities, computed using euro-denominated swap rates combined with the Ultimate Forward Rate (UFR) as discount rates.[7] Second, the required funding ratio reflects the fund's risk profile. Resembling the value-at-risk constraint, it is defined as the level of assets needed to ensure that, with 97.5% probability, the funding ratio does not fall below 100% within 1 year. Interest rate risk and equity risk are the dominant components, so a higher required funding ratio implies that the asset allocation carries greater risk and therefore necessitates a larger buffer.[8]

The minimum funding ratio is 104.2%, and plans with funding ratios below 110% are not allowed to adjust their pension payments for inflation. The allowed level of indexation then increases monotonically, and only plans with funding ratios above 140% are allowed to offer full indexation. In addition, if a pension fund does not meet its minimum or required funding requirements, it must submit a recovery plan to the regulator and is prohibited from taking on additional risk. The reason for these high funding requirements is that a Dutch pension sponsor has no obligation to provide financial aid to its pension plan if it becomes underfunded.

[6] Corporate pension funds are set up by individual firms, industry-wide pension funds cover specific industries or sectors (e.g., civil servants), and professional-group pension funds cover workers within a particular profession (e.g., veterinarians or pharmacists).

[7] More precisely, zero-coupon rates are extracted from swap rates and converted into forward rates. For maturities above 20 years, the discount curve is a weighted average of market-implied forward rates and the UFR. See section 2 of Jansen (2025) for details.

[8] We provide details and sample calculations for the required funding ratio in Internet Appendix IA.1.

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By contrast, pension funds in other jurisdictions (e.g., the United States or Japan) can be substantially underfunded, but in these instances, the pension sponsor is ultimately liable for the pension liabilities.

**1.1.2 Pension risk management.** Pension risk management involves two components. First, the board of the pension fund sets the strategic asset allocation, taking into account the risk–return profile of the pension assets. For example, the board may require the fund to allocate 50% of its assets to fixed income securities and the remaining 50% to equities. Second, because pension funds face interest rate risk due to the long duration of their liabilities, the board also determines the amount of interest rate risk the fund is permitted to take. These risk-taking decisions are typically revisited every 3 years. Importantly, adjustments to risk taking are restricted when a pension fund faces a funding gap, that is, when its funding ratio lies below the required funding ratio.

The long duration of the liabilities exposes pension funds to substantial interest rate risk. Because the present value of the pension liabilities is calculated using swap rates, a drop in the swap rate increases pension liabilities. Unless the value of the pension assets increases by the same amount, a drop in interest rates deteriorates the funding status of the plan. Throughout this paper, we use the term "duration" to capture the sensitivity of assets and liabilities to changes in interest rates.

The pension fund can hedge this sensitivity to interest rates in three ways. First, it can invest the majority of its assets in long-dated bonds and thereby match the duration of its assets and liabilities.[9] The drawback of this approach is that long-dated bonds have substantially lower expected returns than equities, which reduces the probability that the fund can provide full indexation of pension payments. Second, the fund could in principle use repurchase agreements to take a levered position in fixed income securities while still allocating a large share of its portfolio to equities. While British pension funds are permitted to use direct leverage (e.g., Andersson 2023), the Dutch law effectively prohibits the use of repo transactions for strategic leverage and limits them to meeting temporary liquidity needs.[10] This regulation makes Dutch pension funds an ideal laboratory to study the liquidity risk introduced through derivatives usage only. Third, the pension fund can take a fixed-receiver position in interest rate swaps to match the duration of its liabilities without using direct

[9] As we lay out in Internet Appendix IA.1, pension regulation imposes that the duration of equities equals zero. Hence, pension funds must use fixed income securities to hedge the duration risk of their liabilities. We later show that our results remain unchanged when assuming a positive duration for equities.

[10] The law prohibits pension funds from straight borrowing. They may lend out bonds to meet temporary liquidity needs, but any cash obtained in this way must eventually be backed by asset sales if the need persists.



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balance-sheet leverage. Although pension funds could combine these approaches to hedge their interest rate risk, we show below that interest rate swaps are their preferred hedging tool.

Hedging interest rate risk with swaps offers two key advantages for pension funds. First, interest rate swaps allow pension funds to obtain duration exposure without purchasing long-term bonds, as the notional amount is not exchanged and only margin must be posted. Second, hedging with swaps has the same effect on the required funding ratio as purchasing safe euro-denominated bonds. In practice, as we demonstrate in the following section, pension funds do not fully hedge the interest rate risk of their liabilities. Consequently, most pension funds exhibit a duration gap, with liabilities having a longer duration than assets.

In addition to hedging the duration risk of their liabilities, pension funds also invest in bonds with a range of maturities. These investments are critical for matching the cash flows from their liabilities (Leibowitz 1986; Fabozzi, Tong, and Zhu 1990; Jansen 2025). In line with the notion that pension funds have long-dated liabilities, pension funds face large cash flows from liabilities with more than 15 years to maturity (see Figure IA.1 in the Internet Appendix). At the same time, pension funds also have significant liabilities with cash flows due within the next 7 years. Accordingly, pension funds invest in medium-term bonds (maturities between 1 and 7 years) to match these shorter-term obligations.

### 1.2 Interest rate swaps and margin requirements

Entering into a swap position requires pension funds to post initial margin when the contract is initiated and variation margin as the mark-to-market value of the swap evolves (BCBS and IOSCO 2015). The initial margin is a fixed amount calibrated to cover potential future losses in the event of counterparty default, whereas the variation margin reflects the daily mark-to-market change of the position and often exceeds the initial margin by several orders of magnitude (e.g., Czech et al. 2021). When interest rates rise, the mark-to-market value of a fixed-receiver swap declines, and pension funds must post additional variation margin. We refer to these increases in variation margin as margin calls. Margin calls are particularly severe when collateral must be posted in cash, because it can force pension funds to liquidate parts of their asset portfolio.[11] As a result, sharp increases in interest rates expose pension funds that use swaps for duration hedging to liquidity risk whenever margin requirements exceed their cash holdings. We illustrate this mechanism in a numerical example in Internet Appendix IA.2.1.

Since 2012, the European Market Infrastructure Regulation (EMIR) has required European pension funds to exchange collateral on their

[11] If pension funds receive cash margin, they invest this cash according to their strategic asset allocation.

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derivative positions (European Parliament and Council 2012). Until June 2023, pension funds were exempt from mandatory central clearing and could choose the type of collateral posted in their bilateral OTC derivative positions (ESMA 2022). Central clearing requires all counterparties to post cash as variation margin. Even though clearing was not mandatory during our sample period, most bilateral swaps were nonetheless collateralized in cash (ISDA 2017). Dealers prefer cash collateral because it allows them to hedge their positions through central clearing. This preference can lead to discriminatory pricing in bilateral swaps (Cenedese, Ranaldo, and Vasios 2020) and gives pension funds an incentive to use centrally cleared contracts. In addition, Czech et al. (2021) highlight that the introduction of the leverage ratio further increased banks' preference for cash collateral.

### 1.3 Data

We obtain detailed information on the liabilities, assets, and derivative usage of all Dutch pension funds from the Dutch Central Bank ("De Nederlandsche Bank;" henceforth, DNB). Granular data on all three balance sheet components are rare and make the Dutch pension system an ideal laboratory for studying the feedback effects of pension funds' derivative usage on their asset allocation.

Pension funds report quarterly to DNB and began providing detailed information on their derivative positions in the first quarter of 2012. We drop funds that report for fewer than four consecutive quarters and focus on the period from 2012Q1 to 2022Q4. Our final sample consists of 255 distinct pension funds.

Pension funds directly report the duration of their bond portfolios in years. We compute the contribution of the bond portfolio to total portfolio duration in years as the allocation to fixed income multiplied by the reported bond duration, which is equivalent to the dollar duration of the bond portfolio scaled by AUM. For swaps, as detailed in Internet Appendix IA.3, we compute swap duration as the dollar duration scaled by total AUM, expressed in years. This measure is directly comparable to the duration of the bond portfolio. Hence, a swap (bond) duration of 5 years implies that for an interest rate change $\Delta r$, total assets change by $-5\times\Delta r\times AUM$. To reduce the influence of outliers, we winsorize the level and changes in bond, swap, and total portfolio durations at the 1% level.

Table 1 reports summary statistics for our sample. The average pension fund has €6.7 billion in AUM and allocates 56% of its portfolio to bonds and 33% to equities. The average funding ratio is 110%, with substantial cross-sectional variation (5th to 95th percentile: 92% to 133%). The average required funding ratio is 117% (5th to 95th percentile: 109% to 125%). Net contributions, defined as pension contributions minus pension payments,



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equal on average 0.26% of AUM per quarter.[12] Pension funds have an average liability duration of 19 years and a lower average portfolio duration of 10.7 years. Examining asset and swap durations separately reveals that swaps account for about 60% of the median portfolio duration (= 6.2/10.3 × 100), as measured using our dollar duration measure. The asset duration of 4.3 years reflects the pension funds' average bond allocation of 56% and average bond portfolio duration of 7.8 years.

In addition to these quarterly data, DNB collects monthly information on the asset holdings of the largest pension funds. This sample contains 42 pension funds but covers between 85% and 90% of the total AUM of Dutch pension funds. Internet Appendix Table IA.3 shows that, apart from size, the pension funds in this holdings sample exhibit characteristics similar to those of the full population of Dutch pension funds. Asset holdings do not "look through" underlying mutual fund investments, except for the two largest pension funds. For these funds, following Jansen (2025), we use their known shares in mutual funds together with mutual funds' reported holdings to recover the underlying indirect holdings.[13] Panel B of Table 1 provides summary statistics of the monthly holdings and shows that pension funds invest the majority (66%) of their fixed income portfolios in government debt, of which 53% is allocated to German and Dutch government bonds.

Finally, we identify periods in which pension funds faced realized margin calls on their swap positions as follows. We classify a month as a margin call event if the largest 10-day change in the 20-year swap rate exceeds 40 basis points. Because pension funds report asset holdings at a monthly frequency, our analysis focuses on the months in which these large increases occur. Panel C of Table 1 lists the 8 months with such significant rate surges.[14] The panel also reports the corresponding 10-day and monthly changes in the 20-year swap rate, as well as the monthly returns of the Morgan Stanley Capital International (MSCI) Index and the European Investment Grade Bond Index. In addition, we present the average bid–ask spreads of German and Dutch government bonds across different maturity buckets during these months.

### 1.4 Stylized facts

We now present three stylized facts that motivate our analysis. First, pension funds hold large fixed-receiver positions in euro-denominated interest rate swaps. Second, whereas the duration of their bond portfolios

[12] Pension contributions and payments are reported annually. We assume they remain constant throughout the year and divide by four to obtain quarterly net contributions.

[13] After this adjustment, Jansen (2025) shows that the fraction of direct assets relative to total assets equals 85% on average.

[14] May and June 2015; March 2020; and April, June, August, September, and December 2022.



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**Table 1**
**Summary statistics**

A. Quarterly regulatory data

| | Mean | SD | P5 | P50 | P95 |
|---|---|---|---|---|---|
| *AUM* (bln) | 6.74 | 34.02 | 0.06 | 0.75 | 24.35 |
| *Funding ratio* | 110.17 | 14.47 | 92.30 | 108.56 | 133.00 |
| *Required funding ratio* | 116.92 | 5.74 | 109.10 | 116.50 | 124.80 |
| *Net contributions (% AUM)* | 0.26 | 0.83 | −0.74 | 0.10 | 1.69 |
| *Allocation fixed income* | 55.70 | 21.30 | 19.63 | 56.15 | 87.18 |
| *Allocation cash* | 2.26 | 4.69 | −0.36 | 1.12 | 8.62 |
| *Allocation equity* | 33.31 | 12.23 | 16.21 | 32.04 | 53.89 |
| *Duration liabilities* | 19.38 | 3.67 | 13.95 | 19.00 | 25.89 |
| *Duration portfolio* | 10.65 | 4.17 | 4.40 | 10.30 | 17.70 |
| *Duration bonds* | 7.76 | 3.28 | 2.40 | 7.30 | 13.60 |
| *Duration assets* | 4.29 | 2.23 | 1.09 | 4.00 | 8.44 |
| *Duration swaps* | 6.53 | 4.46 | 0.00 | 6.18 | 14.39 |

B. Monthly bond holdings data

| | Mean | SD | P5 | P50 | P95 |
|---|---|---|---|---|---|
| *% Euro area* | 70.04 | 18.52 | 35.91 | 70.77 | 100.00 |
| *% Corporate bonds* | 34.07 | 20.78 | 0.00 | 34.06 | 69.09 |
| *% Government bonds* | 65.93 | 20.78 | 30.91 | 65.94 | 100.00 |
| *% in Germany* | 31.67 | 18.55 | 5.51 | 29.34 | 66.66 |
| *% in Netherlands* | 20.86 | 18.97 | 0.00 | 18.43 | 58.62 |

C. Rate surge months

| | $\Delta r^{Swap}$ (%) | | *Return* (%) | | *Bid–Ask* (bp) | | |
|---|---|---|---|---|---|---|---|
| | 10d | 1m | MSCI | IG | <1y | 1–7y | >7y |
| May 2015 | 0.55 | 0.28 | 0.1 | - 0.5 | 3.7 | 4.8 | 12.0 |
| Jun 2015 | 0.44 | 0.40 | −2.5 | - 1.9 | 3.8 | 5.0 | 14.9 |
| Mar 2020 | 0.45 | 0.11 | −13.5 | - 3.5 | 5.2 | 7.8 | 15.8 |
| Apr 2022 | 0.42 | 0.54 | −8.4 | - 3.2 | 3.5 | 6.8 | 14.3 |
| Jun 2022 | 0.59 | 0.32 | −8.8 | - 2.5 | 4.1 | 7.3 | 18.3 |
| Aug 2022 | 0.55 | 0.62 | −4.3 | - 5.7 | 4.7 | 9.6 | 17.8 |
| Sep 2022 | 0.50 | 0.51 | −9.5 | - 4.5 | 4.8 | 9.8 | 20.2 |
| Dec 2022 | 0.53 | 0.47 | −4.3 | - 3.9 | 7.6 | 9.8 | 27.1 |

Panel A shows *AUM* (billions); *Funding ratio* (pp); *Required funding ratio* (pp); *Net contributions* (pp); *Allocation fixed income* (pp); *Allocation cash* (pp); *Allocation equity* (pp); and *Duration liabilities, Duration portfolio, Duration bonds, Duration assets*, and *Duration swaps* (years). Cash includes MMFs as well as (safe) short-term bonds and (term) deposits with initial maturities of less than 1 year. Panel B reports summary statistics on the bond holdings data to geographies and bond types (% total bond holdings). Panel A is based on quarterly data from 2012Q1–2022Q4, and panel B is based on monthly data from 2012M1–2022M12. Panel C shows the 10-day change and monthly change in the 20-year swap rate during months in which 10-day changes in the 20-year swap rate exceed 40 basis points (defined as rate surge months). Panel C also shows the monthly returns of the MSCI Index and the European Investment Grade Bond Index in each of the months, as well as the average bid–ask spreads of German and Dutch government bonds with different maturities (in basis points).

remains relatively stable over time, the duration of their swap positions fluctuates substantially. Third, pension funds' cash buffers are too small to absorb the variation margin required when interest rates rise, and for most funds the resulting margin calls exceed not only their available cash but also the improvement in funding ratios that accompanies higher interest rates.



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**1.4.1 Swap usage by Dutch pension funds.** The Dutch pension sector relies extensively on interest rate swaps. In our sample, the share of pension funds using interest rate swaps increased from 75% in 2012 to 95% in 2022; for the 42 largest pension funds, swap usage is essentially 100% throughout the sample period. Illustrating the importance of Dutch pension funds in the swap market, Figure 1 panel A reports the net notional of outstanding euro-denominated swaps in 2021Q4 across Dutch counterparties. Dutch pension funds are net fixed-rate receivers, with receiver positions that exceed those of Dutch banks and vastly exceed the swap usage of Dutch insurance companies and other domestic counterparties. To put the numbers from Figure 1 into perspective, we estimate that Dutch pension funds held around 34% of the European noncentrally cleared swap market, and approximately 4% of the total European swap market.[15]

Panel B of Figure 1 illustrates the scale of Dutch pension funds' swap positions using market values. Between 2021Q4 and 2022Q3, the total market value of their swap portfolios fell from €58.2 billion to −€81.9 billion. Over the same period, the 20-year swap rate increased by 2.3%, implying a dollar duration of about € 6.09 trillion.[16] In other words, for a 1-percentage-point change in interest rates, the value of the swap portfolio changes by approximately €60.9 billion, or around 3% of total Dutch pension assets.[17] This estimate is close to the implied dollar duration from panel A, where the total net notional amount of fixed-receiver swaps equals € 357 billion. Assuming a swap duration of 20 years, this corresponds to a dollar duration of roughly € 7.14 trillion.

**1.4.2 Interest rate hedging and swap usage.** Figure 2 shows the median liability duration and the median total portfolio duration of the pension funds in our sample over time, together with the 25th and 75th percentiles. The median liability duration is approximately 19 years, whereas the median portfolio duration—combining the duration of assets and derivatives—fluctuates around 10.5 years. This implies that the median pension fund maintains a duration gap of roughly 8.5 years, with

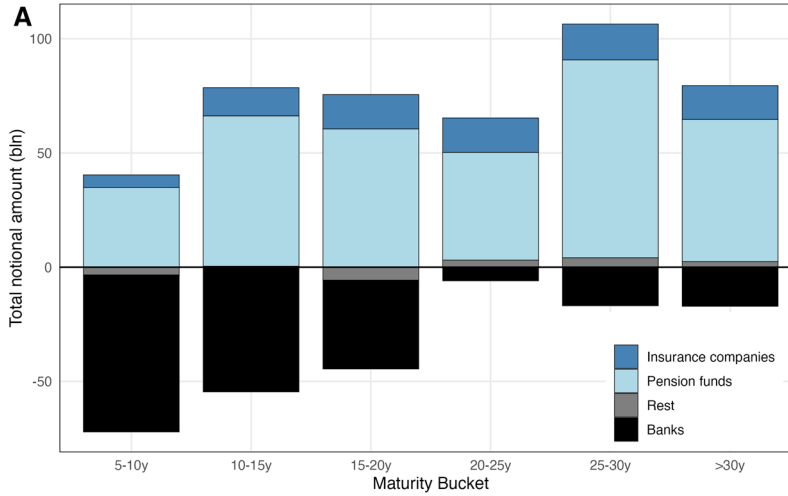
[15] This number is based on back-of-the-envelope calculations using data from the Bank for International Settlements (https://www.bis.org/statistics/derstats.htm). In 2021Q2, the total gross notional amount of all outstanding interest rate derivatives contracts was $475 trillion. Of this amount, 83% were interest rate swaps, 24% had a maturity above 5 years, 26% were euro-denominated, and 12% were not held by central counterparties. Multiplying these fractions suggests the total size of the European swap market with maturities longer than 5 years was around $24.6 trillion. Of this amount, $2.9 trillion was held by institutions other than central counterparties. Using our data, we find that the gross notional of interest rate swaps with more than 5 years to maturity held by Dutch pension funds amounted to $986 billion in 2021Q4.

[16] The implied dollar duration is an approximation, because it assumes pension funds do not alter their swap positions between 2021Q4 and 2022Q3.

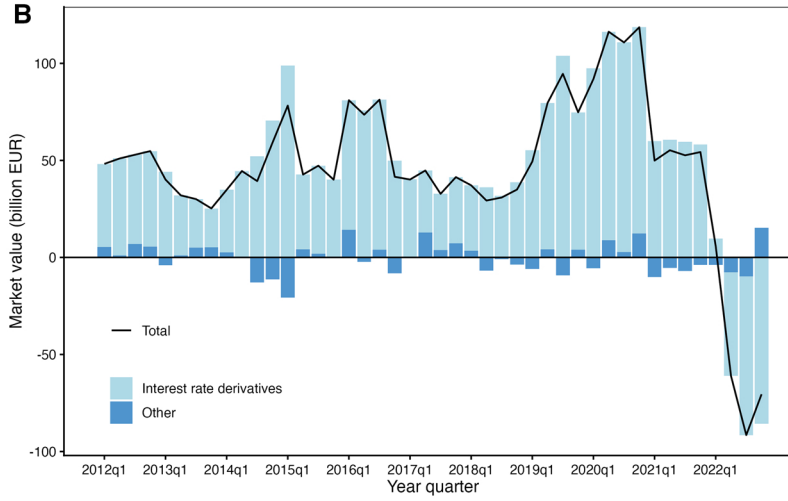
[17] To obtain € 6.09 trillion, we first compute the total change in the present value of swaps as $58.2-(-81.9)=140.1$. Dividing by 2.3% yields the dollar duration.



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Net notional Euribor swaps Dutch counterparties




Market value interest rate swaps and other derivatives

**Figure 1**
**The size of pension funds in the swap market**
Panel A shows the net notional amount in Euribor interest rate swaps for different maturity buckets for banks, pension funds, insurance companies, and the rest established in the Netherlands. The data source is the EMIR database, and calculations are based on the average week-day positions in 2021Q4. Panel B shows the mark-to-market value of interest rate swaps and other derivatives held by Dutch pension funds. Source: DNB statistics.

liabilities substantially longer in duration than assets. The bottom line in Figure 2 plots the median asset duration, computed as the duration of the portfolio excluding derivatives. In contrast to total portfolio duration, asset



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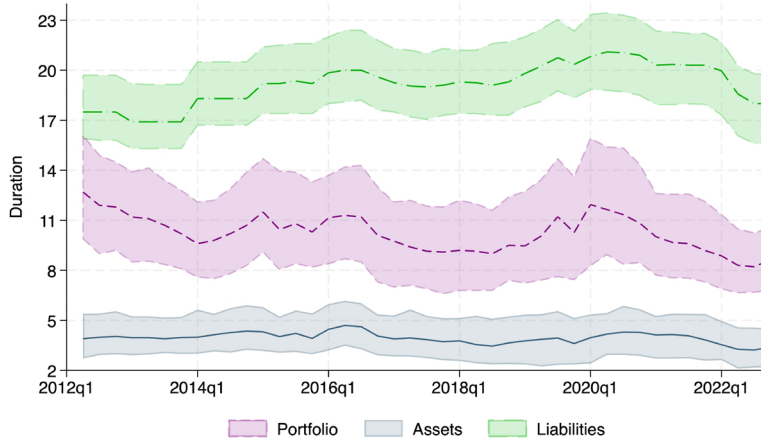



**Figure 2**
**Duration of pension funds' liabilities and portfolios**
This graph shows the 25th percentile, median, and 75th percentile of the duration of the liabilities (green dashed line), long-only assets (blue solid line), and total portfolio (purple dashed line) over time. Total portfolio duration equals the sum of asset and swap duration. The quarterly sample period is 2012Q1–2022Q4.

duration is virtually constant at around 4 years.[18] The figure also reveals substantial cross-sectional variation in swap duration across pension funds, in contrast to the relative stability of asset duration. Taken together, Figure 2 indicates that pension funds adjust their overall portfolio duration primarily through interest rate swaps while leaving the duration of their long-only bond portfolios largely unchanged.

To further illustrate that swap positions are used for hedging, Internet Appendix Figure IA.2 shows that pension funds' funding ratios would exhibit substantially higher volatility had they not used interest rate swaps. This finding is consistent with Sen (2023), who documents a similar role for swaps in hedging interest rate risk for U.S. insurance companies.

**1.4.3 Pension funds face substantial margin call risk.** We now illustrate how using interest rate swaps exposes pension funds to liquidity risk when interest rates increase. To that end, we construct a proxy for realized margin calls. For each pension fund, we compute the dollar duration of its swap portfolio relative to AUM and multiply this value by the percentage-point change in the 20-year swap rate during the 8 episodes of substantial interest rate increases defined in Section 1.3.

[18] To put this asset duration into perspective, recall that the average pension fund invests roughly 50% of its assets in bonds. Because equities carry zero duration under the regulatory framework, the total asset duration is approximately half the duration of the bond portfolio, which averages around 8 years. We illustrate the cross-sectional average fixed-income allocation and fixed income duration in Figure IA.9 in the Internet Appendix.

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Panel A of Figure 3 shows the distribution of realized margin calls during periods of interest rate surges. The average realized margin call equals 3% of AUM and even exceeds 6% for 10% of our sample. To put these numbers into perspective, panel B shows the distribution of realized margin calls adjusted for cash holdings as a fraction of total assets.[19] For 62% of the pension funds in our sample, the realized margin calls exceed their available cash buffers. Internet Appendix Figure IA.3 further shows that margin call risk is unrelated to pension funds' lagged cash positions, suggesting that funds do not appear to manage their ex ante cash holdings in anticipation of margin requirements.

As of 2018, pension funds directly report the margin calls they would receive when interest rates rise to the regulator. We illustrate the close link between our estimates and the reported numbers in the Internet Appendix (Figure IA.4).[20] Throughout the paper, we show that our results remain similar when using reported margin calls instead of our proxy (despite losing the early years of our sample period).

To shed more light on the magnitude and implications of margin call risk, recall from Section 1.3 that most pension funds have a negative duration gap between their assets and liabilities. Thus, rising interest rates typically improve the funding status of pension funds because they lower the present value of pension liabilities more than the present value of pension assets. To put margin calls from derivatives into perspective relative to these potential improvements in funding status, we compute changes in funding ratios as follows:

$$\Delta FR_{i,t} = \frac{A_{i,t} - L_{i,t} - \left(A_{i,t-1} - L_{i,t-1}\right)}{A_{i,t-1}}, \tag{1}$$

where $A_{i,t}$ equals the AUM of pension fund $i$ at time $t$, and $L_{i,t}$ denotes the corresponding value of the liabilities. We then compare $\Delta FR_{i,t}$ to margin call risk (also taken relative to AUM) and show in Figure 4 that margin call risk outweighs the improvement in funding ratios in 72% of the cases.[21] Moreover, the correlation between $\Delta FR_{i,t}$ and margin call risk is weak. This is because a pension fund's funding status can change due to changes in interest rates, if interest rate risk is not fully hedged, and due to stock returns. By contrast, margin calls materialize when interest rates increase and are not affected by stock returns. Phrased differently, large negative equity returns often coincide with rate surge episodes and

[19] Cash holdings include MMFs, as well as (safe) short-term bonds and (term) deposits with initial maturities of less than 1 year.

[20] The reported margin calls tend to be lower than our proxy because some funds have buffers against margin calls in the form of cash holdings in a margin account.

[21] This finding is quantitatively similar if we instead use the reported margin calls.



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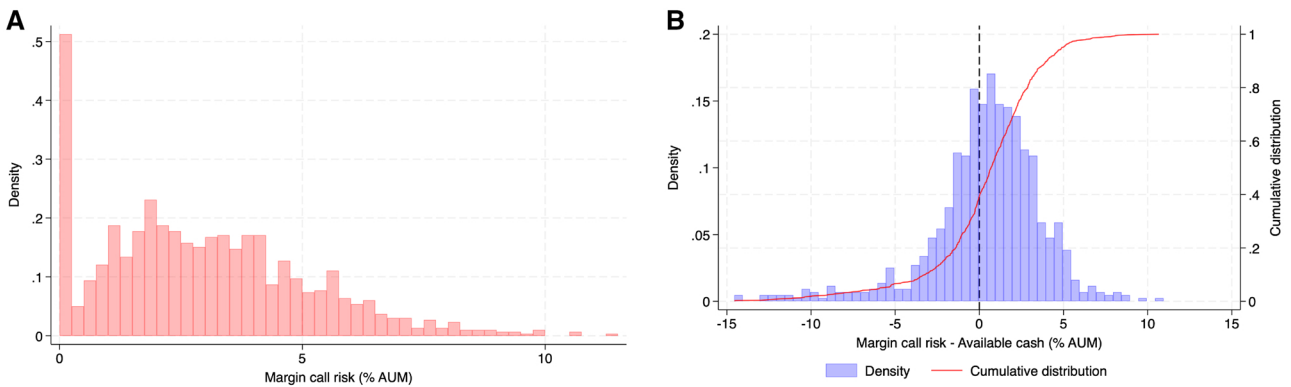



**Figure 3**
**Margin call risk versus available cash**
Panel A shows the distribution of realized margin calls as a fraction of AUM for each pension fund during periods of interest rate surges (2015M5, 2015M6, 2020M3, 2022M4, 2022M6, 2022M8, 2022M9, and 2022M12). Panel B shows the (cumulative) distribution of realized margin calls minus available cash relative to AUM for each pension fund during periods of interest rate surges. The realized margin call risk equals the lagged swap duration times the maximum realized 10-day change in the 20-year swap rate. Available cash includes MMFs, as well as (safe) short-term bonds and (term) deposits with initial maturities of less than 1 year.



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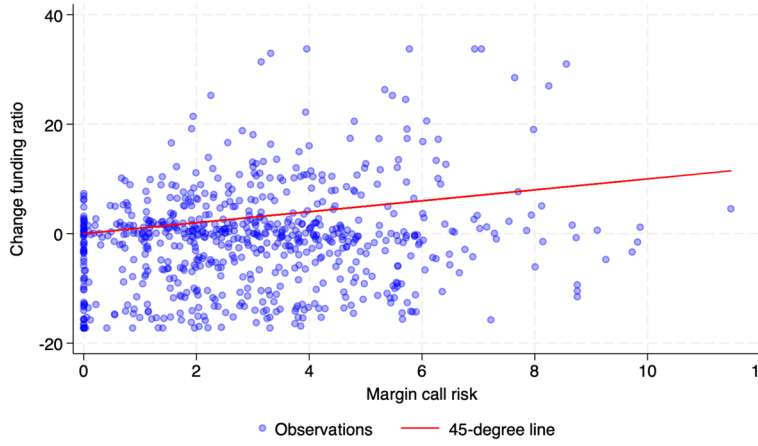



**Figure 4**
**Margin call risk versus changes in funding ratios**
This figure shows a scatter plot of the realized margin calls and the changes in funding status, both as a fraction of AUM, for each pension fund during periods of interest rate surges: 2015M5, 2015M6, 2020M3, 2022M4, 2022M6, 2022M8, 2022M9, and 2022M12. The realized margin call risk equals the lagged swap duration times the maximum realized 10-day change in the 20-year swap rate. The changes in funding ratios are computed as the change in the assets minus liabilities from the quarter starting prior to the swap rate rise to the quarter after.

offset the liability-driven improvement in funding status. As documented in Section 1.3, during episodes of substantial rate hikes the average monthly return on the MSCI Index equals −6.4%, while the average return on the European Investment Grade Bond Index equals −3.2%.

## 2. Hypotheses

Building on the institutional background and stylized facts from Section 1, we now derive three hypotheses to guide our empirical analysis. The idea behind our hypotheses is as follows. Pension funds with lower funding ratios face tighter portfolio constraints because they are not permitted to use direct leverage. Interest rate swaps allow these funds to synthetically increase their portfolio duration without violating these direct leverage constraints. However, swap usage also exposes pension funds to the risk of large margin calls when interest rates rise, potentially forcing them to liquidate parts of their asset holdings.

To derive our hypotheses, consider a pension fund with assets $A$ and liabilities $L$. The fund allocates a fraction $s$ of its assets to stocks, a fraction $b$ to a bond portfolio with the same duration $D$ as its liabilities, and holds the remaining assets $1-s-b$ in cash. In addition, it can engage in interest rate swaps with duration $D$ as fixed receiver, and we denote the notional amount invested in receiver swaps by $N = nA$. In these receiver swaps, the pension fund exchanges variable short-term rates for fixed rates, and the



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resulting cash flows are identical to those of a levered position in the bond portfolio. While pension funds are prohibited from taking levered positions in the bond market, swaps have an initial value of zero and therefore offer an indirect way of obtaining leverage.

To keep the development of our hypotheses focused on interest rate swaps, we take the allocations to stocks and bonds as given and discuss the optimal allocation in Internet Appendix IA.4. We further set the duration of stocks and cash holdings to zero.[22] Since pension funds typically view their asset allocation relative to their liabilities, they incorporate the liability risk directly into their portfolio decisions (e.g., Redington 1952; Sharpe and Tint 1990; Hoevenaars et al. 2008; van Binsbergen and Brandt 2016; Jansen 2025). Hence, we assume that the fund focuses on its funding status $F := A-L$ and uses interest rate swaps to hedge the risk of declining interest rates.

The pension fund's risk management aims to cap the adverse effect of a change $\Delta r$ in interest rates on the funding status at $E = eA$. Using a linear approximation, we obtain

$$[(b+n)A-L]D\Delta r \geq -eA. \tag{2}$$

Solving for $n$ yields

$$n \geq \frac{L}{A} - b - \frac{e}{D\Delta r}. \tag{3}$$

Equation (3) shows that $L/A-b$ is the swap exposure required to fully hedge the duration risk of liabilities.[23] In the more general case, the required swap exposure may be lower if the fund is willing to accept larger funding status losses (larger $e$) when rates decline. Nevertheless, in all cases the required swap exposure is decreasing in the funding ratio as funds with lower funding ratios must use more swaps to limit the sensitivity of their funding status to interest rate changes. This observation leads to our first hypothesis.

> **Hypothesis 1**. *Pension funds with lower funding ratios ($A/L$) use more swaps.*

In our framework, using more swaps translates into a higher dollar duration of the swap portfolio. An alternative explanation might be that pension funds use swaps to enhance their returns by taking more risk, thereby engaging in "reaching for yield" or risk-seeking behavior

[22] Setting the duration of equities to zero is motivated by pension regulation. Later in the paper, we examine how a positive equity duration would affect our results.

[23] Note that this is the duration gap before accounting for swaps. As pension funds hedge more aggressively, the residual gap, that is, the difference between liability duration and total portfolio duration including swaps, falls.



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(e.g., Lian, Ma, and Wang 2019; Lu et al. 2019). However, as we show in the empirical analysis, swap usage is driven primarily by liability-driven interest rate hedging rather than yield enhancement, making such alternative motives unlikely.

We now turn to the implications of swap usage, which are that an increase in interest rates lowers the value of a fixed-receiver position. When the resulting loss exceeds the pension fund's cash holdings, the fund receives a margin call. To minimize the cost of meeting such a margin call, the fund should sell its safest and most liquid assets first (e.g., Brown, Carlin, and Lobo 2010). Because (safe) bond sales typically incur lower transaction costs than stock sales (e.g., Chordia, Sarkar, and Subrahmanyam 2005), and because rate surge episodes tend to coincide with large negative equity returns (Table 1), we expect pension funds to respond to rising swap rates by liquidating parts of their bond holdings. The amount of assets the fund must sell (as a fraction of total assets) in response to a margin call is given by

$$\max(nD\Delta r-(1-s-b),\,0). \tag{4}$$

Hence, pension funds with longer swap durations are more susceptible to receiving larger margin calls and therefore more likely to reduce their bond holdings when interest rates rise.

> **Hypothesis 2**. *Increases in swap rates coincide with reductions in bond holdings. This effect is more pronounced for pension funds with longer swap durations.*

Our setting mitigates three common concerns. First, changes in bond holdings may be correlated with changes in interest rates because of portfolio rebalancing. However, rising interest rates correspond to lower bond prices, and pension funds not facing margin calls would therefore increase their bond holdings to maintain their strategic asset allocation. Hence, reductions in bond holdings when swap rates rise cannot be explained by rebalancing. Moreover, while rebalancing toward strategic weights typically occurs gradually and with a lag, margin calls must be met within the same month.

Second, the duration gap between liabilities and assets narrows when interest rates rise because the liability duration drops more than the asset duration due to a convexity effect. Although this channel could also prompt reductions in bond holdings if pension funds target stable duration gaps, we show in a numerical example in Internet Appendix IA.2.2 that it primarily affects long-term bonds and may even induce purchases of short-term bonds, which is inconsistent with our findings as we show later. As with strategic rebalancing, such adjustments do not necessarily occur immediately, particularly since a smaller duration gap implies lower interest rate risk and thus a higher funding status.



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Third, changes in swap rates could be endogenous to pension funds' asset reallocations. Incorporating pension funds' swap positions into our analysis mitigates this concern. If, for example, pension funds' reductions in fixed income allocations were pushing swap rates up, this effect would operate at the aggregate level and affect all funds equally, so we would not observe a stronger relationship for funds with larger swap exposures.

We examine Hypothesis 2 further and assume the pension fund holds a portfolio of bonds with varying degrees of liquidity and risk. In the presence of a realized margin call, we expect pension funds to mitigate its impact by selling their most liquid bond holdings first. This behavior is consistent with evidence that investors respond to liquidity shocks by liquidating their most liquid assets (e.g., Scholes 2000; Coval and Stafford 2007; Ben-David, Franzoni, and Moussawi 2012; Manconi, Massa, and Yasuda 2012; Ma, Xiao, and Zeng 2022). Safe government bonds typically exhibit lower transaction costs (e.g., Chordia, Sarkar, and Subrahmanyam 2005), smaller liquidity premia (Meyer, Reinhart, and Trebesch 2022), and lower price volatility, with the latter generally associated with greater market liquidity (e.g., Brunnermeier and Pedersen 2009). Hence, when pension funds are forced to liquidate part of their portfolios to meet margin calls, we expect them to sell their safest and most liquid bond holdings first.

> **Hypothesis 3**. *During periods of sharp increases in swap rates, pension funds sell their safest and most liquid bond holdings first.*

To test Hypothesis 3, we assume that medium-term government bonds from safe issuers constitute the most liquid segment of pension funds' fixed income portfolios (e.g., O'Sullivan and Papavassiliou 2020). This assumption is supported by the summary statistics in panel C of Table 1, which show significantly lower bid–ask spreads for government bonds with maturities below 7 years. Importantly, unlike short-term bonds, medium-term government bonds are not cash-like assets. As discussed in Section 1, pension funds also hold these securities to match the cash flows from their medium-term liabilities (Figure IA.1).

### 2.1 Consequences of margin calls

Pension funds are large investors in sovereign bond markets. As noted by Coval and Stafford (2007); Chang, Hong, and Liskovich (2015); and Da et al. (2018), selling pressure by large institutional investors can affect the prices of the securities they trade. Hence, it is plausible that bond sales by pension funds have an adverse price impact. In particular, if Hypothesis 3 holds, our setting implies that the liquidation of safe medium-term government bonds during periods of substantial swap rate increases should depress their prices.

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## 3. Pension Funds' Swap Usage

In this section, we first test Hypothesis 1 by examining the link between pension funds' swap usage and their hedging demand. We then discuss alternative explanations for the observed patterns.

### 3.1 Swap usage and hedging demand

Figure 5 presents bin-scatter plots of lagged funding ratios against (i) total portfolio durations and (ii) swap durations. Panel A shows that pension funds with lower lagged funding ratios hold portfolios with longer durations than funds with higher funding ratios.[24] Panel B demonstrates that this relationship is essentially unchanged when focusing on swap durations instead of total portfolio durations, indicating that the negative link between funding ratios and portfolio duration is entirely driven by swap positions. Taken together, these patterns provide suggestive evidence in support of Hypothesis 1.

We next study the relationship between portfolio duration ($Dur_{i,t}$) and the lagged funding ratio ($FR_{i,t-1}$) using panel regressions of the following form:[25]

$$Dur_{i,t} = \beta FR_{i,t-1} + \gamma C_{i,t-1} + \alpha_i + \lambda_t + \varepsilon_{i,t}, \tag{5}$$

where $C_{i,t-1}$ denotes a vector of time-varying fund-level controls. These include the lagged duration of pension liabilities, the lagged logarithm of fund AUM, lagged net contributions (as a percentage of AUM), and the lagged required funding ratio.

Column 1 of Table 2, panel A, confirms the motivating evidence from Figure 5: funds with lower funding ratios hold portfolios with longer durations. Column 2 shows that this relationship remains robust when including fund and time fixed effects. To interpret the economic magnitude, recall from Table 1 that the standard deviation of the funding ratio is 14.47%. Hence, the estimate in column 2 implies that a 1-standard-deviation increase in the funding ratio is associated with a reduction in portfolio duration of approximately 0.56 years ($=-3.853\times0.1447$).

A critical question for our analysis is whether the longer portfolio durations of more constrained pension funds are driven by their swap positions or by their bond positions. To examine this, we replace portfolio duration with swap duration on the left-hand side of Equation (5). This test corroborates the suggestive evidence from Figure 5 that the overall

[24] Figure IA.5 in the Internet Appendix illustrates the strong negative relation between the cross-sectional average funding ratio and swap durations over time.

[25] We cluster the standard errors at the fund level to allow for within-fund correlation. The coefficient on the lagged funding ratio remains statistically significant when we double-cluster at the fund and time levels, but because our time-series dimension is relatively short, our baseline specification does not cluster at the time level (Angrist and Pischke 2009).



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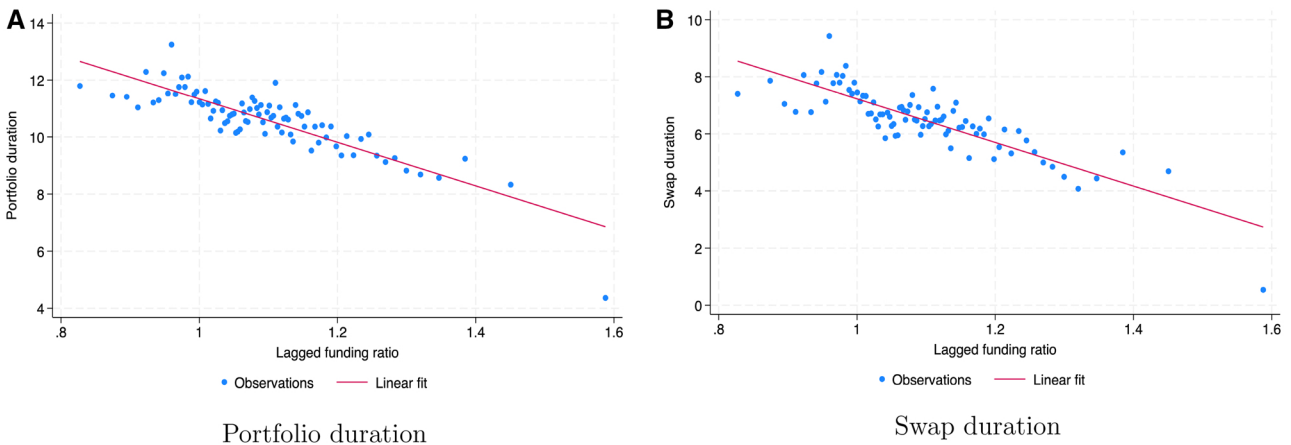



**Figure 5**
**Portfolio duration and lagged funding ratio**
This figure shows bin-scatter plots of the lagged funding ratio against the total portfolio duration (panel A) and against the swap duration (panel B). The total portfolio duration equals the sum of the asset and swap duration. Each dot represents a group of pension fund-quarter observations, whereby we split pension fund-quarter observations in $\sqrt{N} = 89$ groups. The durations are in years and the quarterly sample period is 2012Q1–2022Q4.



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**Table 2**
**Funding constraints and swap usage**

A. Duration in levels

| | *Portfolio duration* | | *Swap duration* | | *Asset duration* | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Funding ratio* $(t-1)$ | −2.806* | −3.853** | −3.479** | −4.072** | 0.511 | −0.323 |
| | [1.575] | [1.811] | [1.480] | [1.903] | [0.568] | [1.020] |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Time FE | No | Yes | No | Yes | No | Yes |
| Fund FE | No | Yes | No | Yes | No | Yes |
| Obs. | 7,366 | 7,366 | 7,178 | 7,177 | 7,375 | 7,375 |
| $R^2$ | 0.308 | 0.782 | 0.252 | 0.779 | 0.245 | 0.745 |

B. Duration in changes

| | *Portfolio duration* | | *Swap duration* | | *Asset duration* | |
|---|---|---|---|---|---|---|
| | OLS | IV | OLS | IV | OLS | IV |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Δ *Funding ratio* | −6.635*** | −9.520*** | −5.823*** | −8.790*** | −0.771*** | −0.468 |
| | [0.431] | [0.662] | [0.421] | [0.747] | [0.136] | [0.354] |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Obs. | 7,345 | 7,345 | 7,098 | 7,098 | 7,375 | 7,375 |
| $R^2$ | 0.134 | 0.124 | 0.114 | 0.103 | 0.287 | 0.287 |
| Coeff. first stage | | 0.722 | | 0.715 | | 0.723 |
| *t*-stat | | 36.97 | | 36.15 | | 37.05 |

Panel A regresses the level of *Portfolio duration*, *Swap duration*, and *Asset duration* on the lagged level *Funding ratio*. Controls include the lagged *Liability duration, Log AUM, Net contributions*, and *Required funding ratio*. Fund and time fixed effects are included as indicated. Panel B presents first difference regressions of changes in *Portfolio duration, Swap duration*, and *Asset duration* on contemporaneous changes in *Funding ratio*. We report the results for both OLS and IV regressions. For the IV regressions, we instrument the changes in *Funding ratio* by the lagged equity allocation times the return on the MSCI Index. Controls include the changes in *Liability duration, Log AUM, Net contributions, Required funding ratio*, and *10-year German yield*. The critical value of the *t*-stat in Stock, Wright, and Yogo (2002) for rejecting weak instruments equals 4.05. Standard errors are clustered at the fund level and reported in brackets. The quarterly sample period is 2012Q1–2022Q4. $^{*}p < .1$; $^{**}p < .05$; $^{***}$p < .01

effect is primarily driven by swaps. Column 3 of Table 2 shows that the link between lagged funding ratios and swap duration is statistically significant and similar in magnitude to the relationship in column 1. When adding fund and time fixed effects, column 4 indicates that a 1-standard-deviation decrease in the funding ratio increases swap duration by approximately 0.59 years ($=-4.072\times0.1447$). By contrast, columns 5 and 6 show no meaningful relationship between lagged funding ratios and the duration of the asset portfolio.

Instead of examining the link between lagged funding ratios and swap usage in levels, we can also study the relation in first differences. A challenge in this setting is that OLS estimates in first differences may suffer from reverse causality. For example, if a pension fund increases its swap duration immediately before an interest rate decline, its funding ratio will improve by more than that of a fund whose swap duration remains unchanged. To address this concern, we instrument changes in the funding ratio using the plans' allocation to equities interacted with the return on



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the MSCI World Index, which provides exogenous variation in asset values. More specifically, we estimate regressions of the form

$$\Delta Dur_{i,t} = \beta \widehat{\Delta FR}_{i,t} + \gamma \Delta C_{i,t} + \varepsilon_{i,t}, \tag{6}$$

where $\widehat{\Delta FR}_{i,t}$ denotes the instrumented change in the funding ratio. Because the instrument is highly correlated across funds, including time fixed effects would absorb a substantial part of the identifying variation. Instead, we control for changes in 10-year German government bond yields to capture changes in duration driven by the level of interest rates. Furthermore, first-differencing eliminates fund-specific effects, so fund fixed effects are not included in this specification.

Our instrument satisfies the exclusion restriction if its realizations affect portfolio duration only through their effect on the funding ratio. A potential violation would arise if pension funds systematically adjusted the desired duration of their portfolios in response to stock market returns themselves. To assess whether this concern is relevant, we compare the piecewise correlation between changes in portfolio duration and the instrument for two subsamples: observations with small changes in the funding ratio ($|\Delta FR| < 1\%$) and those with large changes ($|\Delta FR| > 5.5\%$). For the subsample with small $\Delta FR$, the correlation is –0.007 ($p$-value = 0.803); for the subsample with large $\Delta FR$, it is –0.313 ($p$-value = 0.000). This pattern is inconsistent with an expectations channel because pension funds do not systematically adjust their portfolio duration in response to high or low equity returns in situations where the equity return has no material effect on the funding ratio.[26]

Panel B of Table 2 reports the results of both the OLS and IV regressions in first differences. Consistent with the findings in panel A, we observe that pension funds increase their portfolio duration when funding ratios decline. As before, this effect is entirely driven by adjustments in the swap portfolio. In terms of economic magnitude, the IV estimate in column 4 implies that a 10-percentage-point decrease in the funding ratio raises the duration of the swap portfolio by approximately 0.88 years.

In the Internet Appendix (Table IA.4), we show that our results are robust to using the lagged funding gap instead of the funding ratio in the regression specification. The funding gap provides a direct and arguably more accurate measure of the regulatory constraints faced by pension funds (see Section 1.1). However, given the high correlation of −0.95 between the funding ratio and the funding gap, it is not surprising that the estimates remain qualitatively and quantitatively similar.

To conclude our analysis, we show that pension funds with worse funding ratios do not hold larger cash buffers. This finding is consistent

[26] For example, a low (high) equity return can have a muted effect on funding ratios if simultaneously bond or other asset class returns are high (low).



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with the theory of Ang, Chen, and Sundaresan (2012), who demonstrate that underfunded pension funds allocate their available capital to equities to maximize the probability of becoming fully funded in the future. Hence, the larger swap positions observed for pension funds with a weaker funding status expose them to greater liquidity risk when margin calls arise. We illustrate this point with a bin-scatter plot in the Internet Appendix (Figure IA.6).[27]

### 3.2 Alternative explanations: Equity duration and risk taking

In this section, we first consider whether a positive equity duration could affect our results. We then discuss the alternative view that swap usage may reflect additional risk taking.

Although pension regulation assigns a duration of zero to equities, pension funds may perceive equities as sensitive to interest rate movements and could therefore use equity investments for economic interest rate hedging. Such a view is consistent with evidence that stock returns and bond yields have been positively correlated during parts of our sample period (e.g., Campbell, Pflueger, and Viceira 2020). Hence, even though we find that pension funds do not increase their bond durations when funding ratios decline, they may instead increase their effective equity duration. To examine this possibility, Table 3 modifies regression (5) by adding a fund's stock duration to the measures of portfolio and asset duration.[28] We define stock duration as the equity allocation of fund $i$ multiplied by the average duration of the equity market, which we set to 40 years based on van Binsbergen (2026).[29]

Table 3 shows that the results are virtually identical to those in Table 2. In particular, the coefficient on the lagged funding ratio remains statistically insignificant when asset duration includes equity duration. This suggests that pension funds do not use equity positions to dynamically hedge their interest rate risk. Consistent with Sen (2023) and Jansen (2025), our results indicate that pension funds' hedging decisions are primarily driven by regulatory, rather than economic, hedging incentives.

An alternative interpretation of pension funds' swap usage is that larger swap positions reflect risk-seeking behavior rather than hedging motives. Risk taking with derivatives has been documented in the banking sector

[27] One potential concern is that pension funds may not sharply distinguish between cash-like assets and medium-term bonds. We therefore repeat the analysis underlying Figure IA.6, replacing cash holdings with safe medium-term bond holdings. Consistent with the cash results, Figure IA.7 shows that pension funds with worse funding ratios tend to hold fewer safe medium-term bonds.

[28] We do not use the IV specification in Equation (6) here because both the instrument and equity duration are functions of the fund's equity allocation.

[29] van Binsbergen (2026) estimates market-implied stock durations between 33 and 50 years for the S&P 500 Index and between 32 and 42 years for the Eurostoxx Index. Our results are robust to using shorter or longer values within this range.



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**Table 3**
**Funding constraints and swap usage: Equity duration**

| | *Portfolio duration* | | *Swap duration* | | *Asset duration* | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Funding ratio* $(t-1)$ | −2.862* | −4.408** | −3.479** | −4.072** | 2.116 | −0.474 |
| | [1.591] | [2.101] | [1.480] | [1.903] | [1.492] | [1.872] |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Time FE | No | Yes | No | Yes | No | Yes |
| Fund FE | No | Yes | No | Yes | No | Yes |
| Obs. | 7,359 | 7,359 | 7,178 | 7,177 | 7,375 | 7,375 |
| $R^2$ | 0.677 | 0.841 | 0.252 | 0.779 | 0.268 | 0.700 |

We regress *Portfolio duration, Swap duration*, and *Asset duration* on the lagged *Funding ratio*. We add to *Portfolio duration* and *Asset duration* the implied equity duration, defined as the allocation to equity times the market implied equity duration of 40 years. Controls include the lagged *Liability duration, Log AUM, Net contributions*, and the lagged *Required funding ratio*. Fund and time fixed effects are included as indicated. Standard errors are clustered at the fund level and reported in brackets. The quarterly sample period is 2012Q1–2022Q4. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

(e.g., Begenau, Piazzesi, and Schneider 2015). However, our evidence is inconsistent with the view that pension funds use swaps for risk taking for three reasons.

First, as shown in Figure IA.2, funding ratios would be more volatile in the absence of swaps—precisely the opposite of what risk-seeking behavior would imply. Second, if pension funds were using swaps to speculate on falling interest rates, they should simultaneously increase the duration of their bond portfolios. However, we find no empirical relationship between funding ratios and asset durations, indicating that funds do not use derivatives to take directional interest rate positions. Third, pension regulation prohibits underfunded funds from increasing their equity risk. Thus, they cannot use swaps to maintain duration exposure while reallocating fixed income assets toward riskier securities.

Taken together, these findings point to liability-driven hedging as the primary motive for swap usage, consistent with the evidence for insurance companies in Domanski, Shin, and Sushko (2017) and Ozdagli and Wang (2019).

## 4. Response to Margin Calls from Swaps

We now turn to testing Hypotheses 2 and 3. To proxy for margin calls, we use changes in swap rates and proceed in three steps. First, to test Hypothesis 2, we examine how changes in swap rates affect pension funds' fixed income holdings. Second, to test Hypothesis 3, we focus on large interest rate surges and study the selling behavior of the most liquid fixed income securities in response to realized margin calls. Finally, we analyze the selling of other asset classes during these rate surges and discuss alternative interpretations.

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### 4.1 Feedback from interest rate swaps to portfolio allocation

We begin with the first part of Hypothesis 2, examining how changes in swap rates affect pension fund asset allocations. Throughout this section, we measure interest rate changes using the maximum 10-day increase in the 20-year swap rate within a given month (based on rolling changes over ten business days). In the Internet Appendix (Table IA.5), we show that our results are virtually unchanged when using monthly swap rate changes instead. As noted in Section 1.3, monthly security-level information is available for 42 large pension funds.

We estimate panel regressions of the following form:[30]

$$\frac{Net\ Buys_{i,t}}{AUM_{i,t-1}} = \beta_0 \Delta r_t^{Swap} + \beta_1 \left( \Delta r_t^{Swap} \times Dur_{i,t-1}^{Swap} \right) + \beta_2 Dur_{i,t-1}^{Swap} + \gamma C_{i,t-1} + \alpha_i + \varepsilon_{i,t}, \tag{7}$$

where $\frac{Net\ Buys_{i,t}}{AUM_{i,t-1}}$ denotes net purchases (buys minus sales) of all fixed income assets by fund $i$ in month $t$, scaled by lagged AUM. $\Delta r_t^{Swap}$ is the maximum 10-day change in the 20-year swap rate, and $Dur_{i,t-1}^{Swap}$ is fund $i$'s lagged swap duration (measured as the dollar duration divided by AUM), capturing exposure to margin call risk. The vector $C_{i,t-1}$ includes the same lagged fund-level controls used earlier (the funding ratio, net contributions, required funding ratio, and liability duration).

To account for rebalancing effects, we further control for the relative performance of a fund's fixed income and equity portfolios. For each fund $i$, we compute the hypothetical change in the fixed income weight assuming no trades:

$$\Delta w_{i,t}^{FI,hyp} = \frac{w_{i,t-1}^{FI}(1+r_t^{IG})}{w_{i,t-1}^{FI}(1+r_t^{IG}) + (1-w_{i,t-1}^{FI})(1+r_t^{MSCI})} - w_{i,t-1}^{FI}, \tag{8}$$

where $w_{i,t-1}^{FI}$ is the lagged fixed income allocation, $r_t^{IG}$ is the return on the European Investment Grade Bond Index, and $r_t^{MSCI}$ is the return on the MSCI Index. A positive value of $\Delta w_{i,t}^{FI,hyp}$ indicates that bonds outperformed stocks and hence a mechanical increase in the fixed income weight. To allow for gradual rebalancing, we include the contemporaneous and the 1- and 2-month lags of $\Delta w_{i,t}^{FI,hyp}$. Including additional lags does not affect the results because higher lags are statistically insignificant.

[30] Because the analysis operates at monthly frequency—with a substantially longer time-series dimension—we cluster standard errors at both the fund and time level. This allows for correlation in the error terms within funds over time and across funds within a given month.



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Column 1 of Table 4 shows that rising swap rates are associated with net selling of pension funds' bond positions. A 1-standard-deviation increase in the swap rate change (0.13%) corresponds to a 0.11% reduction in fixed income holdings. To obtain additional insight, we replace the continuous change in the 20-year swap rate with an indicator $\mathbb{1}\{\text{High}_t\}$ that equals one when the 20-year swap rate rises above its 90th percentile in month $t$. As column 5 shows, the economic effect is more pronounced in these months: during sharp rate increases, pension funds' fixed income allocations fall by 0.30%. This behavior is inconsistent with mechanical rebalancing under strategic asset allocation rules, which would imply increases in fixed income allocations when rates rise (and bond prices fall). Instead, it indicates that pension funds sell fixed income securities in response to rising swap rates, consistent with liquidity needs from margin calls.

When contrasting our results with strategic rebalancing, it is crucial to emphasize the difference in timing. Margin calls must be met immediately, implying asset sales within the same month. In contrast, rebalancing following market price movements typically occurs gradually and with a lag.[31] Consistent with this distinction—and controlling explicitly for rebalancing—Table 4 shows that the contemporaneous hypothetical change in the fixed income allocation is insignificant, whereas its lagged values are negative and statistically significant. This pattern indicates that when equity returns exceed bond returns, pension funds increase their fixed income holdings in subsequent periods to restore their strategic allocation.

We next turn to the second part of Hypothesis 2 and augment the specification with an interaction term between changes in swap rates and a fund's lagged swap duration. The idea behind this test is that funds with larger swap durations face greater exposure to margin call risk, so swap rate changes should have a stronger effect on their bond portfolios. As we discuss further below, this cross-sectional pattern also allows us to separate the margin call channel from the duration rebalancing channel.

Consistent with this prediction, columns 2 and 6 show that the interaction term is statistically significant with the expected sign. By contrast, neither the swap rate change itself nor the indicator for interest rate surges is individually significant. These cross-sectional patterns mitigate the concern that swap rate changes merely reflect pension funds' own rebalancing activities because the effects arise precisely where margin call exposure is greatest.

To address the concern that selling behavior may reflect changes in pension funds' interest rate expectations, we next control for the lagged bond duration of a fund's fixed income portfolio ($Dur^{Bond}_{i,t-1}$) and its

[31] Similarly, rebalancing to adjust for convexity effects between assets and liabilities does not necessarily occur immediately.



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**Table 4**
**Bond holdings and maximum 10-day changes in swap rates**

| | *Net buys / AUM* | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| $\Delta r^S$ | −0.851** | −0.011 | 0.311 | | | | | |
| | [0.355] | [0.617] | [0.910] | | | | | |
| $Dur^S(t-1)\times\Delta r^S$ | | −0.157* | −0.168** | −0.223** | | | | |
| | | [0.081] | [0.089] | [0.092] | | | | |
| $Dur^B(t-1)\times\Delta r^S$ | | | −0.061 | −0.145 | | | | |
| | | | [0.169] | [0.217] | | | | |
| $\mathbb{1}\{\text{High}\}$ | | | | | −0.296* | 0.180 | 0.289 | |
| | | | | | [0.153] | [0.267] | [0.371] | |
| $Dur^S(t-1)\times\mathbb{1}\{\text{High}\}$ | | | | | | −0.086** | −0.088** | −0.111*** |
| | | | | | | [0.034] | [0.033] | [0.034] |
| $Dur^B(t-1)\times\mathbb{1}\{\text{High}\}$ | | | | | | | -0.022 | -0.060 |
| | | | | | | | [0.060] | [0.078] |
| $Dur^S(t-1)$ | | 0.026 | 0.049* | 0.039 | | | 0.026 | 0.014 |
| | | [0.026] | [0.026] | [0.025] | | | [0.026] | [0.026] |
| $Dur^B(t-1)$ | | | 0.139** | 0.141*** | | | 0.127*** | 0.123*** |
| | | | [0.052] | [0.047] | | | [0.041] | [0.043] |
| *Rel. bond return* $(t)$ | 0.015 | 0.009 | 0.004 | −0.126 | 0.025 | 0.018 | 0.014 | −0.153 |
| | [0.031] | [0.034] | [0.033] | [0.227] | [0.033] | [0.035] | [0.034] | [0.233] |
| *Rel. bond return* $(t-1)$ | −0.076 | −0.085 | −0.089* | −0.448* | −0.058 | −0.067 | −0.071 | −0.484* |
| | [0.051] | [0.051] | [0.052] | [0.256] | [0.051] | [0.051] | [0.052] | [0.254] |
| *Rel. bond return* $(t-2)$ | −0.084** | −0.091** | −0.092** | −0.747*** | −0.078* | −0.085** | −0.086** | −0.760*** |
| | [0.041] | [0.041] | [0.041] | [0.245] | [0.042] | [0.042] | [0.042] | [0.242] |
| *Funding ratio* $(t-1)$ | 0.005* | 0.001 | 0.003 | −0.004 | 0.006* | 0.002 | 0.003 | -0.006 |
| | [0.003] | [0.005] | [0.005] | [0.012] | [0.003] | [0.005] | [0.005] | [0.012] |
| *Net contributions* $(t-1)$ | 0.278** | 0.097 | 0.112 | −0.165 | 0.277** | 0.107 | 0.120 | −0.158 |
| | [0.132] | [0.207] | [0.189] | [0.209] | [0.131] | [0.200] | [0.187] | [0.204] |
| *Req. funding ratio* $(t-1)$ | −0.000 | 0.013 | 0.029 | 0.037 | −0.001 | 0.007 | 0.024 | 0.036 |
| | [0.008] | [0.020] | [0.022] | [0.026] | [0.008] | [0.018] | [0.022] | [0.026] |
| *Liability duration* $(t-1)$ | −0.018 | −0.122*** | −0.134*** | 0.012 | −0.017 | −0.115*** | −0.128*** | 0.009 |
| | [0.018] | [0.030] | [0.033] | [0.044] | [0.017] | [0.030] | [0.033] | [0.042] |
| Fund FE | No | Yes | Yes | Yes | No | Yes | Yes | Yes |
| Time FE | No | No | No | Yes | No | No | No | Yes |
| Obs. | 4,460 | 4,460 | 4,460 | 4,460 | 4,460 | 4,460 | 4,460 | 4,460 |
| $R^2$ | 0.01 | 0.02 | 0.02 | 0.06 | 0.01 | 0.02 | 0.02 | 0.06 |

This table shows the results of regressing monthly net purchases of bonds relative to AUM on the maximum 10-day change in the 20-year swap rate within the month ($\Delta r^S$ ; column 1), the corresponding swap rate change interacted with the swap duration ($Dur^S$; column 2), and the swap rate change interacted with the swap duration and bond duration ($Dur^B$; columns 3–4). We also run similar regressions using an indicator variable whether the swap rate change is in the 90th percentile of the distribution ($\mathbb{1}\{\text{High}\}$; columns 5–8). Controls include current and lagged changes in hypothetical bond allocations as specified in Equation (8), *Funding ratio, Net contributions, Required funding ratio*, and *Liability duration*. Fund and time fixed effects are included as indicated. The monthly sample period is 2012M1–2022M12. Standard errors are clustered at the fund and time level and reported in brackets. $^*p < .1$; $^{**}p < .05$; $^{***}p < .01$

interaction with swap rate changes ($Dur^{Bond}_{i,t-1}\times\Delta r^{Swap}_t$). Bond duration is a good first-order approximation of the return on pension funds' fixed income portfolio because, as shown in Table 1, most fixed income holdings consist of safe government debt, for which interest rate risk is the dominant return driver. Moreover, including the interaction between bond duration and rate changes helps isolate whether fixed income sales reflect shifts in



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interest rate expectations rather than liquidity needs arising from margin calls.[32]

The intuition is as follows. Suppose a pension fund expects further increases in interest rates and thus lower future bond returns. Also assume these expectations are positively correlated with the size of a pension fund's swap portfolio—for example, because further increases in interest rates result in larger losses on the swap portfolio for pension funds with more substantial swap positions. If this mechanism drives the selling behavior, we would expect a similar selling behavior among pension funds with long bond durations, as these funds are likewise more impacted by further interest rate increases, which would result in larger losses on their fixed income portfolios.

As shown in columns 3 and 7 of Table 4, we find no relationship between a pension fund's bond duration and its sales of fixed income securities when interest rates rise. This evidence rules out the interpretation that our results are driven by expectations of further rate increases rather than by liquidity needs arising from margin calls.

To control for unobservable fluctuations in financial markets or the broader economy, we next add time fixed effects. Because these fixed effects absorb all time-series variation in swap rates, identification comes from the cross-sectional interaction between rate changes and lagged swap durations. As columns 4 and 8 of Table 4 show, the coefficient on the interaction term remains virtually unchanged, confirming that our results are not driven by common time-varying shocks.

To further mitigate the concern that swap rate changes are endogenous to pension funds' rebalancing activities, we instrument the 10-day changes in 20-year swap rates with the effect of MPSs on the 20-year euro OIS rates (Altavilla et al. 2019).[33] The identifying assumption is that MPSs affect pension funds' portfolio decisions only through their effect on swap rates; it is unlikely that MPSs directly change pension funds' bond demand within the same month. These features support both instrument relevance and the exclusion restriction. Table IA.6 in the Internet Appendix shows that the instrument is strong, with first-stage $t$-statistics ranging from 8.94 to 76.53, well above the 4.05 threshold suggested by Stock and Yogo (2005). Consistent with our baseline findings, pension funds reduce their bond holdings when swap rates rise, and this effect is concentrated among funds with large swap positions.

Finally, we consider whether the observed selling patterns could reflect pension funds' efforts to maintain a constant duration gap. Figure IA.10 shows that pension funds with a higher swap duration have a lower

[32] A potential concern is multicollinearity between bond duration and swap duration. However, as documented in Internet Appendix Figure IA.8, the two variables are only mildly negatively correlated.

[33] The MPS data are available from 2013M10, so the IV sample covers 2013M10–2022M12.



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duration gap. Hence, under the duration rebalancing channel, funds with higher swap duration would need to do less rebalancing when rates rise and should therefore sell fewer bonds, yet these are precisely the funds that sell the most bonds. This cross-sectional pattern is therefore inconsistent with duration rebalancing and instead supports the margin call channel.

To further corroborate this conclusion, note that to keep the duration gap unchanged after an interest rate increase, a pension fund could either reduce its overall fixed income allocation while keeping the bond duration fixed or shorten the duration of its bond portfolio while maintaining the fixed income allocation. However, because pension funds typically commit to their strategic asset allocation for multiple years (see Section 1.1.2), it is unlikely that the observed short-run reductions in fixed income holdings reflect such adjustments. While modifying the duration of the fixed income portfolio is theoretically possible, Internet Appendix IA.2.2 provides a numerical illustration showing that the required adjustments are inconsistent with our empirical findings.

### 4.2 Liquid asset sales during interest rate surges

Turning to Hypothesis 3, we examine whether pension funds sell their most liquid securities first when margin calls materialize. As discussed in Section 1.4, we identify margin call episodes as months in which the maximum 10-day increase in the 20-year swap rate exceeds 40 basis points.

Prior to selling bonds, it is plausible that pension funds use their cash buffers to fulfill margin calls. While our monthly data do not provide detailed information on actual cash balances (e.g., bank deposits), we observe monthly holdings of MMFs. MMFs are cash-like instruments and form part of the quarterly cash measure used in Figure 3. We show in the Internet Appendix (Figure IA.11) that pension funds reduce their MMF positions during periods of substantial increases in interest rates. This finding is consistent with Ghio et al. (2023), who document that MMF outflows in March 2020 were driven by investors facing large margin calls on interest rate derivatives. Stress-test evidence also indicates that Dutch pension funds view MMF sales as a key source of short-term liquidity (DNB and AFM 2024).[34] However, MMF holdings are a small portion of pension assets with the median fund holding less than 1.5% of its assets in MMFs. Consequently, MMF sales (and sales of other cash-like assets) are insufficient to meet larger margin calls. Thus, we next focus on bond sales.

Our hypothesis is that pension funds concentrate their sales in the safest and most liquid bonds. To test this prediction, we construct an indicator $\mathbb{1}\{\text{AAA}\}_{j,t}$ that equals one if security $j$ is a euro-denominated government

[34] Pension funds may also obtain short-term liquidity via repo transactions. In our data, outright sales and repo contracts are easily distinguished: a sold bond leaves the balance sheet, whereas a bond pledged as collateral in a repo remains.



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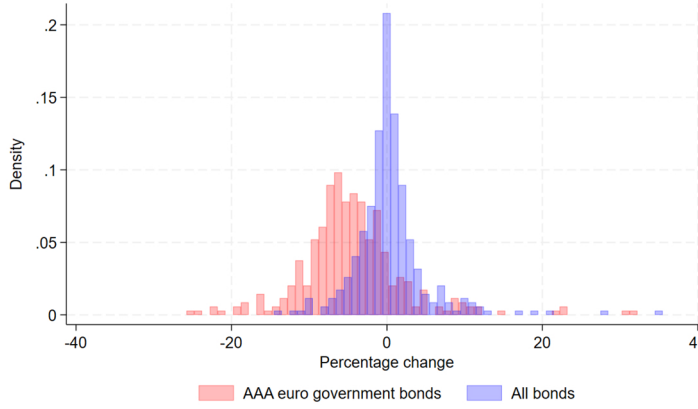



**Figure 6**
**Changes in bond holdings during periods of interest rate surges**
This graph shows the percentage change in nominal bond holdings during periods of interest rate surges—2015M5, 2015M6, 2020M3, 2022M4, 2022M6, 2022M8, 2022M9, and 2022M12—separately for AAA euro-denominated government bonds (red bars) and the total bond portfolio (blue bars).

bond with a triple-A rating from all three major rating agencies and has a remaining maturity exceeding 1 year. We exclude non–euro-denominated safe bonds because margin calls arise from euro-denominated swaps, and liquidating foreign-currency bonds would introduce additional currency risk. During our sample period, only Dutch and German government bonds meet the triple-A criterion (see Table IA.7 in the Internet Appendix). We further exclude bonds with remaining maturity below 1 year to distinguish safe euro-denominated bonds from cash-like assets. Because medium-term bonds are typically more liquid than long-term bonds, we expect sales to concentrate in this segment. We therefore define $\mathbb{1}\{T<7\}_{j,t}$, which equals one if the bond has a remaining maturity below 7 years.

To examine which segments of the fixed income portfolio are liquidated during margin call episodes, we study security-level changes in the notional amounts held by each fund in each of the 8 event months.[35] As a first step, we compare the percentage change in the notional holdings of triple-A euro-denominated bonds with the percentage change in total bond holdings for each fund. As shown in Figure 6, pension funds reduce their positions in triple-A euro-denominated bonds significantly more than in other fixed income securities during these episodes.

We next run panel regressions of the following form:

$$\Delta\%Hold_{i,j,t} = \beta Dur^{Swap}_{i,t-1} \times \mathbb{1}\{\text{AAA}\}_{j,t} \times \mathbb{1}\{T<7\}_{j,t} + \gamma C_{i,j,t} + \delta_{j,t} + \alpha_{i,t} + \varepsilon_{i,j,t}, \quad (9)$$

[35] Using notional changes rather than market-value changes ensures that our results are not mechanically driven by price movements.



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where $\Delta\%Hold_{i,j,t}$ denotes the percentage change in fund $i$'s holding of security $j$ during event month $t$, and $Dur_{i,t-1}^{Swap}$ is the fund's lagged swap duration. The vector $C_{i,j,t}$ contains the two–way interactions between the indicator variables and lagged swap duration separately. The terms $\delta_{j,t}$ are security–time fixed effects, controlling for time-varying differences in bond characteristics (including shifts in the cross-section of expected bond returns), and $\alpha_{i,t}$ are fund–time fixed effects.[36] This specification tests whether pension funds with greater margin call exposure sell more of the most liquid safe bonds during rate surges.

Starting with a specification that interacts only lagged swap duration and the triple-A indicator, column 1 of Table 5 shows that sales of triple-A government bonds are more pronounced for funds with larger swap durations. Column 2 shows that this coefficient is essentially unchanged once we add security–time fixed effects, and column 3 confirms that adding both security–time and fund–time fixed effects leaves our inference unchanged.

To examine whether these sales are concentrated in the most liquid segment of the market, columns 4–6 include the interaction with $\mathbb{1}\{T < 7\}_{j,t}$. The resulting triple interaction term is statistically significant at the 1% level across all specifications, whereas the interaction between lagged swap duration and the triple-A indicator becomes insignificant. In terms of economic magnitude, a 1-standard-deviation increase in swap duration (4.46) reduces each triple-A euro-denominated bond holding with less than 7 years to maturity by $-0.99\%$ ($=4.46\times-0.221$). This pattern is inconsistent with rebalancing aimed at maintaining a constant duration gap as we describe in Internet Appendix IA.2 and is instead consistent with liquidity-driven selling due to margin calls.[37]

We perform three robustness checks for our analysis. First, we repeat the tests using reported margin calls instead of our proxy. As shown in Internet Appendix Table IA.9, the results remain largely unchanged when focusing on reported margin calls and restricting the analysis to the event months for which these data are available.

Second, to address the concern that pension funds may cover margin calls from their cash reserves rather than bond sales, we construct an alternative margin call measure defined as the maximum of zero and realized margin calls (as in Figure 3) net of lagged cash holdings. This net measure isolates the portion of margin calls that funds cannot absorb

[36] Standard errors are clustered at the security level to allow for correlation across pension funds holding the same bond. The coefficient $\beta$ remains statistically significant when we double-cluster at the fund and security levels, which also allows for correlation across securities within a fund.

[37] To further examine the duration gap as potential driver, we test if pension funds rebalance their bond portfolios to short-term bonds (with remaining time to maturity below 1 year). Table IA.8 shows that there are no significant purchases of short-term bonds, which is inconsistent with maintaining a constant duration gap.



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**Table 5**
**Margin call risk and selling of AAA medium-term bonds**

| | % Hold | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $Dur^S(t-1)\times\mathbb{1}\{\text{AAA bond}\}$ | −0.086*** | −0.085*** | −0.080*** | −0.007 | −0.006 | 0 |
| | [0.028] | [0.029] | [0.029] | [0.036] | [0.034] | [0.033] |
| $Dur^S(t-1)\times\mathbb{1}\{\text{AAA bond}\}\times\mathbb{1}\{T<7\}$ | | | | −0.196*** | −0.220*** | −0.221*** |
| | | | | [0.053] | [0.059] | [0.059] |
| $Dur^S(t-1)\times\mathbb{1}\{T<7\}$ | | | | 0.01 | 0.012 | 0.01 |
| | | | | [0.008] | [0.009] | [0.009] |
| $\mathbb{1}\{\text{AAA bond}\}\times\mathbb{1}\{T<7\}$ | | | | 1.426** | | |
| | | | | [0.599] | | |
| $Dur^S(t-1)$ | 0.011*** | 0.037*** | | 0.006 | 0.030** | |
| | [0.004] | [0.010] | | [0.006] | [0.012] | |
| $\mathbb{1}\{\text{AAA bond}\}$ | 0.607* | | | 0.051 | | |
| | [0.321] | | | [0.402] | | |
| $\mathbb{1}\{T<7\}$ | | | | −0.084 | | |
| | | | | [0.114] | | |
| Security FE | No | No | No | No | No | No |
| Security-Time FE | No | Yes | Yes | No | Yes | Yes |
| Fund-Time FE | No | No | Yes | No | No | Yes |
| Obs. | 420,118 | 355,131 | 355,131 | 420,118 | 355,131 | 355,131 |
| $R^2$ | 0.00 | 0.54 | 0.55 | 0.00 | 0.54 | 0.55 |

In columns 1–3, we perform regressions of the percentage change in monthly individual-level nominal bond holdings on the lagged swap duration ($Dur^S$) interacted with a dummy that indicates whether the bond is an AAA-rated government bond, denominated in euro, and with a remaining time to maturity above 1 year during periods of interest rate surges: 2015M5, 2015M6, 2020M3, 2022M4, 2022M6, 2022M8, 2022M9, and 2022M12 ($\mathbb{1}\{\text{AAA bond}\}$). In columns 4–6, we also interact with a dummy that indicates whether the bond has a remaining time to maturity that is below 7 years ($\mathbb{1}\{T<7\}$). Security, security-time, and fund-time fixed effects are included as indicated. Standard errors are clustered at the security level and reported in brackets. $^*p<.1$; $^{**}p<.05$; $^{***}p<.01$

with existing cash holdings and therefore must meet through asset sales. As reported in Table IA.10, our results are robust to this alternative specification. We complement this test with a sample split: Table IA.11 replicates our main analysis separately for pension funds with below-median (panel A) and above-median (panel B) lagged cash holdings. Consistent with the cash buffer narrative, the effect is concentrated among low-cash funds, for which the triple-interaction coefficient ranges from −0.252 to −0.548 across specifications. Among high-cash funds, none of the triple-interaction coefficients are statistically significant, indicating that funds with sufficient cash reserves do not need to liquidate medium-term bonds to meet margin calls to the same extent.[38]

Third, following Section 4.1, we use MPSs from Altavilla et al. (2019) as an instrument for changes in swap rates to address the concern that swap

[38] The average duration gap is nearly identical across the two subsamples (10.09 years for high-cash funds and 9.44 years for low-cash funds), and if anything points against the duration channel: funds with larger duration gaps (which should sell more under duration rebalancing) are precisely those that sell fewer medium-term bonds.

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rate movements may be endogenous to pension funds' selling behavior. Specifically, we reestimate our regressions for the months in which (i) the 20-year swap rate increases sharply and (ii) MPS have a positive effect on the 20-year euro OIS rate. The idea behind focusing on these months is that the rate surges are linked to (surprising) changes in monetary policy. Four months meet these criteria: 2015M6 (7.5 basis points), 2022M4 (1.9 basis points), 2022M9 (2.4 basis points), and 2022M12 (8.5 basis points). This test constitutes a subsample analysis, and we report the results in the Internet Appendix (Table IA.12). Focusing on this subset leaves our main results unchanged.

### 4.3 Other asset sales in response to margin calls?

We next examine pension funds' alternatives to selling medium-term bonds in response to margin calls. We focus on the potential liquidation of less liquid government bonds, illiquid bonds, and equities.

**4.3.1 Long-term government bonds.** We begin with less liquid long-term government bonds. Recall from panel C of Table 1 that the bid-ask spread differential between medium-term and long-term bonds ranges from 7.2 basis points in May 2015 to 17.3 basis points in December 2022. Thus, even though selling medium-term bonds, as we show later, leads to an adverse price impact, it is still a more cost efficient response to margin calls than selling longer dated bonds once transaction costs are taken into account.

**4.3.2 Illiquid bonds.** Second, we examine whether pension funds also sold illiquid bonds in response to margin calls. Such sales could generate even larger price impacts than the liquidation of liquid bonds (Ellul et al. 2022). We investigate this possibility in Internet Appendix Table IA.13. Using bid–ask spreads from Eikon for all bonds in our sample, we classify illiquid bonds as those with bid–ask spreads in the 80th percentile.[39] Following Equation (9), we interact this illiquidity indicator with the lagged swap duration. In contrast to the pattern for liquid bonds, we find no statistically significant sales of illiquid bonds during months with substantial interest rate increases.

**4.3.3 Equities.** Finally, we consider equities. An important qualitative difference across the eight rate-surge episodes concerns the nature of the underlying rate movements. The sharp rate increases in 2022 reflected a persistent shift in monetary policy, whereas the spikes in 2015 and 2020 were transitory. Persistent increases in interest rates lower the value of

[39] The results are qualitatively similar when using the 90th percentile as the cutoff.



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pension funds' fixed income portfolios and therefore create incentives to rebalance by selling equities. Such rebalancing can occur independently of margin calls.

Figure IA.12 in the Internet Appendix illustrates this distinction. Pension funds sell parts of their bond portfolios in all episodes, but equity sales appear only during the sustained rate increases in 2022.[40] These patterns suggest that the equity sales observed in 2022 primarily reflect portfolio rebalancing. Absent margin calls, the proceeds from such equity sales would normally be used to increase bond holdings. In 2022, however, during persistent and sizable increases in interest rates, it is plausible that pension funds diverted part of these equity-sale proceeds toward meeting margin calls. A similar rebalancing channel for covering margin calls is discussed in EIOPA (2023).

Consequently, our findings likely represent a conservative estimate relative to a scenario in which pension funds held fewer equities. For instance, U.K. pension funds allocate substantially less to equities than their Dutch counterparts: 10–15% compared to 33% in our sample of Dutch pension funds.[41] This difference likely contributed to the more pronounced sell-off and price impact in the U.K. gilt market in September and October 2022.[42]

## 5. Price impact of pension funds

Because Dutch pension funds are major investors in the safe euro-denominated government bond market (e.g., Jansen 2025), it is plausible that their bond sales in response to margin calls exert downward pressure on bond prices. We begin by documenting that bonds more exposed to margin call–driven selling experience larger price declines during periods of interest rate surges. Afterwards, we apply a GIV approach to quantify the aggregate effect of margin calls on safe euro government bond yields.

### 5.1 Cross-sectional bond price effects

To test for price impact, we construct a bond-level measure of margin call risk for bond $j$:

$$MC_{j,t} = \sum_{i=1}^{K_{j,t}} Dur_{i,t}^{Swap} \frac{N_{i,j,t}}{\sum_{i=1}^{K_{j,t}} N_{i,j,t}}, \tag{10}$$

[40] We detrend the cumulative distributions of net buys to remove the effect that pension funds grow over time and therefore purchase more assets mechanically.

[41] U.K. equity allocations are from https://www.chicagofed.org/publications/chicago-fed-letter/2023/480; Dutch allocations are from Table 1.

[42] Additionally, the pooled LDI structure in the United Kingdom contributed to the exacerbated sell-off and price impact as well, because this structure slowed recapitalization, effectively forcing LDIs to sell gilts (Pinter, Siriwardane, and Walker 2025).



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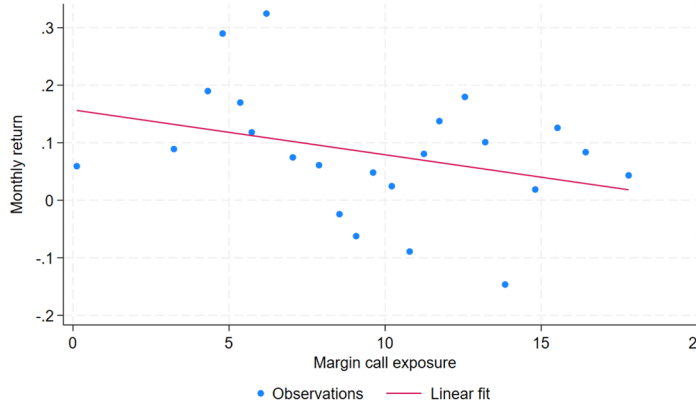



**Figure 7**
**Margin call risk exposure and bond returns**
This graph plots our measure of margin call exposure at the bond-level against the bond returns during the 10-day window of maximum 20-year swap rate changes for bonds with remaining time to maturity below 7 years, orthogonal to the component of returns driven by time to maturity. We include all periods of interest rate surges: 2015M5, 2015M6, 2020M3, 2022M4, 2022M6, 2022M8, 2022M9, and 2022M12. Each dot represents a group of bond-month observations, whereby we split bond-month observations in $N = 23$ groups.

where $Dur_{i,t}^{Swap}$ denotes the swap duration of pension fund $i$ in quarter $t$ (measured as the dollar duration of the swap portfolio, relative to AUM), $N_{i,j,t}$ is the notional amount of bond $j$ held by fund $i$ in month $t$, and $K_{j,t}$ is the number of funds holding bond $j$ at time $t$. Following Ma, Xiao, and Zeng (2022), this measure weights the swap duration of each fund by its notional exposure to the bond, thereby capturing the bond's exposure to margin call risk.

Given the results from the previous section, we focus our price-impact analysis on German and Dutch government bonds with remaining maturities below 7 years.[43] We obtain issuance-level information from auction schedules and complement this with daily (clean) midmarket prices from Thomson Reuters Eikon, from which we compute 10-day returns over the rate-surge windows.

Figure 7 presents a bin-scatter plot of our bond-level margin call risk measure against maturity-adjusted 10-day returns for medium-term bonds ($T < 7$). The figure shows that bonds with higher exposure to margin call risk experience lower returns during periods of sharp rate increases.

To test this relationship formally, we estimate

$$r_{j,t} = \alpha + \beta MC_{j,t-1} + \gamma C_{j,t} + \varepsilon_{j,t}, \tag{11}$$

[43] Focusing on this maturity category also helps isolate the price impact associated with margin calls from potential effects of convexity rebalancing, which primarily affects long-dated bonds.



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where $r_{j,t}$ is the 10-day percentage change in the clean price of bond $j$, $MC_{j,t-1}$ is the margin call risk measure computed in the month prior to the return window, and $C_{j,t}$ is a set of bond-level controls. These controls include time to maturity, its square, bond age, coupon rate, log outstanding volume, an indicator for inflation-linked bonds, and an indicator for Dutch sovereign issuers.

Column 1 of Table 6 shows that bonds with greater exposure to pension funds' margin call risk experience significantly lower 10-day returns. It is likely that for the same selling pressure, the price impact is larger for less liquid bonds, such as longer-term bonds (e.g., O'Sullivan and Papavassiliou 2020), older issuances (e.g., Pasquariello and Vega 2009), smaller issuances (e.g., Helwege and Wang 2021), and bonds issued by the Netherlands (e.g., Papavassiliou and Xia 2025).[44] To account for such differences, we also interact the margin call risk measure with time to maturity, bond age, log outstanding volume, and an indicator variable that equals one if the bond is issued by the Netherlands in column 2.[45] Consistent with our hypothesis, longer-term bonds and older issuances experience a larger price impact when margin call risk materializes. Column 3 demonstrates that the result is robust to including year-to-maturity fixed effects, indicating that our results are robust to controlling for second order effects such as curvature. In terms of economic magnitude, taking the average coefficient of −0.053 across columns 1–3, we find that a 1-standard-deviation increase in margin call risk (=2.1) reduces returns for the average bond by 11 basis points, or about 17% of the average return (=−0.65%) during months with substantial interest rate increases.

As an additional test, we reconstruct $MC_{j,t}$ by replacing $Dur_{i,t}^{Swap}$ in Equation (10) with reported margin calls (available from 2018 onward). Columns 4–6 of Table 6 confirm that the results are robust when using reported instead of estimated margin calls.

We next investigate whether the results are indeed driven by the heterogeneous exposure of bonds to pension funds' margin call risk. To this end, we construct a counterfactual measure that captures the equally weighted swap duration of the pension funds holding bond $j$:

$$CF_{j,t} = \frac{1}{K_{j,t}} \sum_{i=1}^{K_{j,t}} Dur_{i,t}^{Swap}. \tag{12}$$

$CF_{j,t}$ reflects the average swap duration of all pension funds holding a given bond, but it does not necessarily map into the bond's margin call

[44] Papavassiliou and Xia (2025) show that German bonds have lower bid-ask spreads than other euro area countries such as France and Italy. We also show in Figure IA.13 that Dutch bonds have slightly higher bid-ask spreads compared to German bonds over our sample period.

[45] We demean *TTM*, *Age*, and *Log(Outst)* so that we can interpret the coefficient on our margin call risk measure as the average effect.



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**Table 6**
**Price impact of margin call risk**

| | Margin call proxy | | | Margin call reported | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| $MC\,(t-1)$ | -0.012** | −0.058** | −0.088*** | −0.046*** | −0.127*** | −0.132*** |
| | [0.006] | [0.024] | [0.027] | [0.005] | [0.018] | [0.020] |
| *TTM* | −0.137*** | −0.091** | | −0.130*** | −0.094*** | |
| | [0.028] | [0.041] | | [0.022] | [0.021] | |
| $TTM^2$ | −0.019*** | −0.020*** | | −0.020*** | −0.016*** | |
| | [0.005] | [0.005] | | [0.004] | [0.004] | |
| *Age* | −0.017*** | −0.006 | −0.005 | −0.009*** | −0.003 | −0.003 |
| | [0.006] | [0.008] | [0.008] | [0.003] | [0.003] | [0.003] |
| *Coupon* | 0.060*** | 0.070*** | 0.066*** | 0.023* | −0.005 | −0.013 |
| | [0.022] | [0.014] | [0.013] | [0.013] | [0.010] | [0.009] |
| *Log(Outst)* | −0.043** | −0.087** | −0.111*** | −0.027* | 0.015 | 0.022 |
| | [0.017] | [0.042] | [0.041] | [0.015] | [0.016] | [0.017] |
| 𝟙{Inflation-Linked} | −0.246 | −0.237 | −0.233 | −0.338 | −0.409* | −0.410* |
| | [0.203] | [0.183] | [0.175] | [0.210] | [0.211] | [0.212] |
| 𝟙{Dutch} | −0.010 | −0.137* | −0.145* | −0.012 | 0.003 | 0.021 |
| | [0.023] | [0.079] | [0.081] | [0.017] | [0.030] | [0.036] |
| $MC\,(t-1) \times TTM$ | | −0.007* | −0.013*** | | −0.017*** | −0.018*** |
| | | [0.004] | [0.005] | | [0.003] | [0.004] |
| $MC\,(t-1) \times Age$ | | −0.002** | −0.002** | | 0.000 | 0.000 |
| | | [0.001] | [0.001] | | [0.001] | [0.001] |
| $MC\,(t-1) \times Log(Outst)$ | | 0.005 | 0.009 | | −0.012*** | −0.016*** |
| | | [0.006] | [0.006] | | [0.004] | [0.005] |
| $MC\,(t-1) \times$ 𝟙{Dutch} | | 0.020 | 0.021 | | −0.005 | −0.010 |
| | | [0.013] | [0.014] | | [0.008] | [0.009] |
| Year-to-maturity FE | No | No | Yes | No | No | Yes |
| Obs. | 510 | 510 | 510 | 390 | 390 | 390 |
| $R^2$ | 0.53 | 0.53 | 0.54 | 0.55 | 0.56 | 0.56 |

The dependent variable is the 10-day bond return during periods of swap rate surges. The main independent variable is the lagged measure of margin call risk (MC $(t-1)$) as defined in Equation (10), using lagged swap durations as a proxy for margin call risk in columns 1–3 and reported margin calls in columns 4–6. Controls include *TTM*, $TTM^2$ , *Age, Coupon, Log(Outst)*, an indicator whether the bond is inflation-linked (𝟙{Inflation-Linked}), and an indicator whether the bond is issued by the Netherlands (𝟙{Dutch}). In columns 2, 3, 5, and 6 we interact the margin call risk measure with *TTM, Age, Log(Outst)*, and 𝟙{Dutch}. We demean *TTM, Age*, and *Log(Outst)* to interpret the coefficient on MC $(t-1)$ as the average effect. In columns 3 and 6 we also control for year-to-maturity fixed effects. Periods with swap rate surges are 2015M5, 2015M6, 2020M3, 2022M4, 2022M6, 2022M8, 2022M9, and 2022M12. As reported margin calls are only available from 2018 onward, columns 4–6 do not include the first two surge episodes. Standard errors are clustered at the bond-level and reported in brackets. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

exposure. This counterfactual differs materially from $MC_{j,t}$ because of the large cross-sectional variation in both swap exposures and position sizes across pension funds; indeed, the correlation between $CF_{j,t}$ and $MC_{j,t}$ is only 0.55. Consistent with this intuition, Internet Appendix Table IA.14 shows no significant relationship between bond returns and the counterfactual measure.

As margin calls create temporary price pressure on bonds, we next examine whether these price effects subsequently reverse. Identifying reversals is complicated by the fact that rate surges often occur in close succession, making it difficult to define a clean postevent window. To overcome this challenge, we focus on the March 2020 rate surge. This episode involved a sharp but temporary increase in interest rates and offers



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the additional advantage that pension funds were unlikely to engage in strategic rebalancing, allowing a cleaner test for reversal dynamics.

Figure 8 shows the cumulative returns of the bonds that were most exposed (80th percentile of margin call risk distribution) and least exposed (20th percentile of margin call risk distribution) to margin call risk. The figure reveals a clear price reversal after roughly 50 days (1.5 months) and thus underscores that sales because of margin calls are not information-driven sales. This is a faster price reversal than found for U.S. stocks, where Coval and Stafford (2007) find that stock prices revert to prefire-sale levels after approximately 4 months, whereas Edmans, Goldstein, and Jiang (2012) find that prices only fully revert after 2 years. In the U.S. corporate bond market, Ellul, Jotikasthira, and Lundblad (2011) find that prices fully revert 30 weeks (7 months) after fire-sale events. The substantially faster price reversal that we find likely reflects the higher liquidity of safe euro bond markets as opposed to the U.S. stock and corporate bond markets. On the other hand, using financial advisor firms' frequent recommendations to reallocate between equity and bond funds as uninformed demand shocks, Da et al. (2018) find price reversal after 10 days in the Chilean stock market.[46] They argue the faster price reversal compared to the U.S. stock market is likely driven by subsequent changes in the advisory firms' recommendations within a very short time period.

Taken together, our results suggest that margin calls imposed on pension funds force them to sell safe and liquid assets, thereby generating temporary downward pressure on the prices of those assets. This mechanism resembles the liquidity spiral highlighted in Brunnermeier and Pedersen (2009). However, while Brunnermeier and Pedersen (2009) study leveraged speculators pledging risky collateral to finance speculative positions, the pension funds in our setting neither speculate nor use direct leverage.

### 5.2 GIV approach

After demonstrating the effect of bond-specific margin call risk on bond prices, we proceed to estimate the aggregate price sensitivity of safe euro medium-term government bonds to pension fund sales using the GIV approach developed by Gabaix and Koijen (2024). This is a useful addition to the cross-sectional analysis, which removes any common effect on bond yields. To apply the GIV, we use the combined holdings of Dutch and German medium-term debt for each pension fund.

We draw on Ma, Xiao, and Zeng (2022), who apply a similar methodology to U.S. Treasuries. As a first step, we focus on percentage changes in holdings of pension fund $i$ ($\Delta q_{i,t}$).[47] Afterwards, we extract idiosyncratic

[46] In the Chilean foreign exchange market, Aldunate et al. (2025) find no sign of reversal within 10 days after uninformed reallocations to foreign assets.

[47] To limit the influence of outliers, we winsorize these changes at the 1% level.



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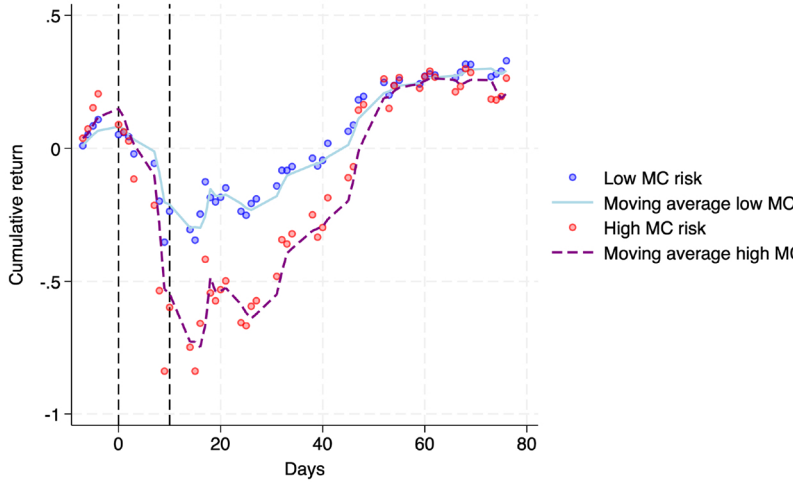



**Figure 8**
**Price reversals**
This figure plots cumulative returns of bonds highly exposed to margin call risk versus bonds with low exposure from March 2 to May 29, 2020. High (low) exposure corresponds to the 80th (20th) percentile of the margin call risk measure in Equation (10). We include Dutch and German government bonds with remaining time to maturity below 7 years. Vertical lines mark the start and end of the 10-day surge window. Returns are orthogonalized with respect to time to maturity, bond age, coupon, outstanding volume, inflation-linked status, and issuer. Plotted returns show the 5-day moving average.

shocks to holdings by estimating the following weighted regressions:[48]

$$\Delta q_{i,t} = \alpha_t + \alpha_i + \beta_{0,i} GDPgrowth_t + \beta_{1,i} \Delta Inflation_t + \beta_{2,i} \Delta Fiscal_t + \beta_{3,i} t + \Delta \check{q}_{i,t}. \quad (13)$$

These regressions include fund- and time-fixed effects. We further control for GDP growth, changes in (CPI) inflation, changes in fiscal capacity (measured as net lending relative to GDP), and a time trend, allowing the coefficients on these macrovariables to vary across funds. To construct the macrocontrols, we take value-weighted averages of the Dutch and German series, using each country's outstanding medium-term government debt as weights.

Because our holdings data are monthly while GDP, inflation, and fiscal capacity are observed quarterly, we allocate one-third of each quarterly growth (or change) to each of the 3 months within quarter $t$. We take the residuals $\Delta \check{q}_{i,t}$ from Equation (13) as the idiosyncratic demand shocks. Our aggregate instrument $Z_t$ is then constructed as the weighted sum of

[48] The weights are $E_i = \frac{\sigma_i^{-2}}{\sum_k \sigma_k^{-2}}$, where $\sigma_i$ denotes the volatility of fund $i$'s holdings.



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these shocks, where each shock is weighted by the share of outstanding medium-term government debt held by fund $i$.[49]

In the second step, we use this instrument to estimate the elasticity $M$ of medium-term government bond yields with respect to a one percent inflow into medium-term government debt. Specifically, we run the following time-series regression:

$$\Delta y_t^3 = MZ_t + \gamma_0 GDPgrowth_t + \gamma_1 \Delta Inflation_t + \gamma_2 \Delta Fiscal_t + \lambda_1 PC_{1,t} + \lambda_2 PC_{2,t} + \lambda_3 PC_{3,t} + \alpha + \varepsilon_t, \quad (14)$$

where $\Delta y_t^3$ denotes the change in the 3-year safe euro government bond yield, computed as the value-weighted average of the Dutch and German 3-year yields, using each country's outstanding medium-term debt as weights. We control for GDP growth, changes in inflation, and changes in fiscal capacity. Additionally, we include the first three principal components of the idiosyncratic changes in holdings to capture common demand comovements that may affect yields.

Table 7 reports the GIV results. Columns 1–5 show that idiosyncratic changes in pension fund holdings provide a strong instrument for identifying the demand-driven component of bond yields. Across specifications, we find that a 1% inflow into the safe euro medium-term government bond market reduces the 3-year yield by approximately 17 basis points. This estimate is stable and robust to controlling for additional principal components.

To gauge the economic magnitude of our findings, we link the GIV estimates to the price pressure resulting from margin calls. For instance, during August 2022, pension funds sold 0.61% of the total amount outstanding of safe euro medium-term debt. Using our GIV estimate, this selling pressure raised the 3-year safe euro yield by 10.4 basis points. Over the same month, the 3-year yield increased by 87 basis points in total, implying that pension-fund sales accounted for roughly 12% of the total rise.

In the Internet Appendix, we perform an additional GIV analysis that broadens market coverage. To expand the set of safe euro bondholders and increase statistical power, we extend our monthly holdings with institution-level data for Dutch insurance companies and banks,[50] as well as monthly holdings of international mutual funds from Morningstar.[51]

[49] Figure IA.14 in the Internet Appendix plots the resulting instrument over time.

[50] Dutch institutions report security-level holdings to DNB on a monthly basis. These data underpin, among other statistics, the Dutch balance of payments, international investment positions, and the financial accounts.

[51] Collectively, these institutions hold, on average, 11% of all outstanding safe euro medium-term debt.



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**Table 7**
**GIV regressions**

| | $\Delta y_t^3$ | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| *Z* | −0.186** | −0.185*** | −0.187** | −0.143* | −0.166** |
| | [0.073] | [0.071] | [0.073] | [0.076] | [0.077] |
| *GDP growth* | | 0.015* | 0.015* | 0.015 | 0.015 |
| | | [0.009] | [0.009] | [0.009] | [0.010] |
| *Inflation (change)* | | 0.258*** | 0.258*** | 0.258*** | 0.258*** |
| | | [0.097] | [0.096] | [0.098] | [0.098] |
| *Fiscal (change)* | | −0.053 | −0.053 | −0.053 | −0.053 |
| | | [0.083] | [0.083] | [0.087] | [0.085] |
| *PC1* | | | 0.015 | 0.015 | 0.015 |
| | | | [0.012] | [0.011] | [0.011] |
| *PC2* | | | | −0.026 | −0.025 |
| | | | | [0.021] | [0.018] |
| *PC3* | | | | | 0.028* |
| | | | | | [0.014] |
| Obs. | 131 | 131 | 131 | 131 | 131 |

This table shows the estimates of the multiplier *M*, which indicates how much yields move when 1% of the safe euro medium-term bond market is sold, using the GIV approach. The dependent variable is the change in the 3-year safe euro bond yield. *Z* is defined as the weighted sum of residual changes in holdings, that is, the GIV. *GDP growth*, changes in *Inflation*, and changes in *Fiscal* are the quarterly GDP growth, (CPI) inflation changes, and fiscal capacity changes, split equally across calendar months. Fiscal capacity is defined as the net amount lent versus borrowed, relative to GDP. *PC1*, *PC2*, and *PC3* are the first three principal components of the residuals in Equation (13). The data are monthly from 2012M2–2022M12. Newey-West standard errors with optimal lags are reported in brackets. $^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

We reestimate the GIV with this extended sample, and Table IA.15 reports coefficients of approximately 13 basis points—close to the estimates obtained in our baseline analysis using only Dutch pension funds' holdings.

Finally, by taking a value-weighted average of the macro variables for Germany and the Netherlands, we effectively treat the two countries as a single economy. However, economic fundamentals differ across the two countries. We therefore repeat the GIV analysis separately for the Netherlands and Germany in the Internet Appendix (Tables IA.16 and IA.17). For the Netherlands, we obtain a coefficient of 13 basis points, which is close to our baseline estimates. For Germany, the average coefficient is 10 basis points and statistically insignificant. One potential explanation is that the German government bond market is more liquid than the Dutch market, which may attenuate price responses to demand shocks. Moreover, the lack of statistical significance for Germany is likely driven by the lower coverage of German bond holdings relative to the Netherlands (see panels C and D of Internet Appendix Figure IA.14).

## 6. Conclusion

Pension funds use interest rate swaps to hedge the interest rate risk of their long-dated liabilities while allocating substantial parts of their portfolios to assets with higher expected returns, such as equities. Although



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this hedging strategy benefits plan participants by mitigating interest rate risk without sacrificing expected returns, our analysis documents a critical downside. The use of interest rate swaps exposes pension funds to significant liquidity risk. When interest rates rise and the present value of swap positions declines, pension funds face margin calls that can exceed 6% of their total AUM. To meet these liquidity demands, they sell safe government bonds, focusing on those with medium-term maturities. This forced selling occurs when bond prices fall, thereby generating an adverse and procyclical price pressure that exposes pension funds to market liquidity risk.

Our findings challenge the traditional view that pension funds are stable, long-horizon investors largely insulated from liquidity risk. Instead, we show that pension liquidity risk arises as a consequence of two regulations designed to enhance financial stability: (i) capital requirements for defined-benefit pension funds and (ii) collateral requirements for interest rate swaps. Since interest rate swaps are a valuable hedging tool for pension funds, one potential policy implication of our analysis is to incorporate capital requirements for liquidity risk and introduce "liquidity stress tests" into pension regulation. These regulations should not curb interest rate swap usage per se, but reduce the liquidity risk associated with using swap contracts and avoid scenarios in which pension funds face margin calls that substantially exceed their available cash positions.

**Code Availability** The replication code is available in the Harvard Dataverse at https://doi.org/10.7910/DVN/PUSRUR.

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